

AM Group

AC-23-237 – Expansion of Additive Manufacturing Processes

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Revision	Date of issue	Reason for revision or original issue
01	XX XXXX	Original issue

Project outline

1. Introduction: Using a broader PSD in PBF-LB to reduce costs.
 - 1a. Previous work
 - 1b. Research impact
 - 1c. Project aims
2. Thicker Layers: Density of specimens built in 120 μm layers.
3. Improving Surface Quality: Developing parameters for 90 μm layers.
4. Building in Coarser powder with a comparison of Standard and Thicker Layers:
 - 4a. Density
 - 4b. PSD, flowability & packing density changes (over time)
 - 4c. Oxygen uptake (over time)
 - 4d. Surface roughness
 - 4e. Microhardness
 - 4f. Tensile properties
 - 4g. Fatigue properties
 - 4h. Gas flow conditions
5. Laser parameter development: Can the spatter out be reduced?
 - 5a. Design of experiments: Lowering the scan speed and the laser power.
 - 5b. Density results.
 - 5c. Surface roughness results.
 - 5d. Microhardness results.
 - 5e. Tensile results.
6. Demonstrator Test Case: Atherton Bikes.
7. Overall conclusions and further work.

1. Introduction: Using a broader PSD in PBF-LB to reduce costs :

1a. Previous work

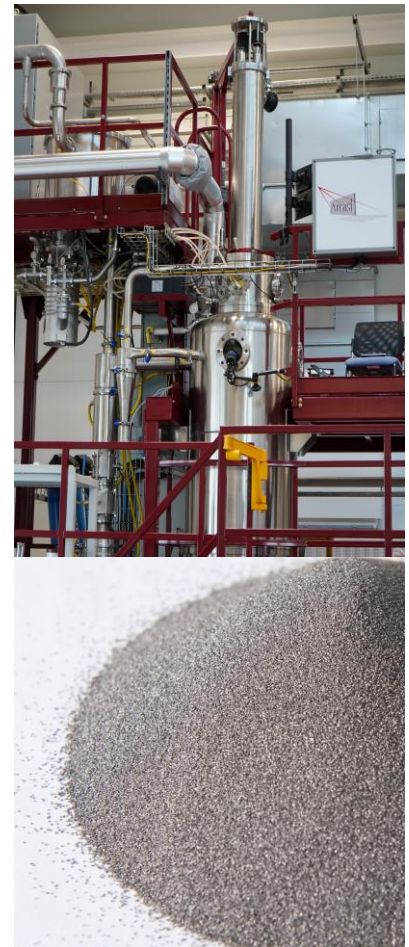
1b. Research impact

1c. Project aims

1. Introduction: High feedstock costs for PBF-LB.

What if we can use a broader range of PSD to build parts?

- For the manufacturing process of laser powder bed fusion (PBF-LB), it is presently recommended for the powder used to be within a narrow particle size range of 15-53 μm to manufacture parts with high resolution for industries including medical and aerospace.
- With the low yields currently achieved by gas atomisation, expanding this range to include larger particles (up to 106 μm) would make use of an underutilised cut of powder, potentially reduce the cost and open-up the process to other industries that may find this manufacturing technology prohibitively expensive to adopt.
- Using an increased powder size distribution (PSD) range in PBF-LB could be employed to decrease manufacturing costs.



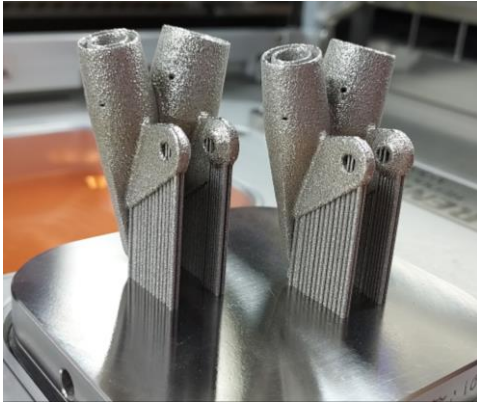
1. Introduction: Using a broader PSD in PBF-LB to reduce costs.

What if we can use a broader range of PSD to build parts?

- A greater yield used from atomization process 15-106 μm up to 40-60 % wt will reduce further processing of underused, larger powder particles.
- Thicker layers (90 μm or 120 μm as opposed to 30 μm and 60 μm) to reduce build times.
- Increased sustainability of material feedstock.
- Overall, reduce the cost per part for AM.

1a. Previous work.

- Increased the PSD from the recommended 15-53 μm to 15-106 μm to build specimens in layers 120 μm thick on the RBV.
- Developed a set of parameters to build specimens up to 99.95% in density.



- Limited mechanical data obtained showed no reduction in yield strength or elongation, however, appeared to reduce the microhardness.
- Reduced size demonstrator parts built in coarser powder built with a poor surface finish.

1b. Research impact.

- Literature review: thicker layers in LPBF-AM has been explored somewhat, but there seems to be less published on work using the broader range of powder sizes.
- Work done on different materials and different platforms focusing on the refinement of laser parameters to obtain fully dense samples, with some examining the microstructure and oxide production in spatter.

Reference	Primary Findings	PSD	LT	Material	Conclusion	Research Gap
Brown and Taylor (2019)	Lower production costs by using coarser powders	45-100 μm	60 μm	Titanium Alloys	Coarser powders reduce manufacturing costs without significantly affecting quality. - Suggests potential for broader industrial application.	Examination of the long-term reliability of coarser powders.
Green et al. (2020)	Optimization of energy density leads to reduced defects	10-45 μm	50-100 μm	Ti-6Al-4V	Defect reduction through optimized energy density. - Improved part density and mechanical properties.	Exploration of energy density in other alloys.
Jones and Lee (2021)	Coarser powders reduce costs without affecting quality	45-100 μm	60-120 μm	Titanium Alloys	Cost-effective manufacturing with Coarser powders. - potential for broader industrial application without quality compromise.	Long-term reliability of coarser powders.
Kumar et al. (2018)	Microstructures affect mechanical properties in LPBF Ti-6Al-4V	10-25 μm	30-90 μm	Ti-6Al-4V	Fine microstructures provide superior strength; columnar grains enhance durability	Detailed analysis of microstructural effects on fatigue life.
Buhairi et al. (2023)	Role of volumetric energy density in morphology and properties	20-60 μm	30-100 μm	Ti6Al4V	High energy density crucial for achieving desired microstructures and mechanical properties	Impact of varied energy densities on different alloys.
Fernandez and Gomez (2021)	Effects of particle size on surface roughness	30-60 μm	50-100 μm	Titanium	Coarser particles associated with increased surface roughness but faster build rates	Fine-tuning particle size for optimal surface finish
Thompson et al. (2022)	Thermal effects on material properties during LPBF	30-70 μm	40-80 μm	Titanium	Thermal management is crucial for achieving desired material properties in LPBF.	Study of thermal effects across various build orientations.

1c. Project aims.

Scaling up (using a production AM machine (Ren500Q) full bed).

- Can the integrated sieve and cyclone system cope with the increased particle sizes?
- Will build location affect density and mechanical performance?
- Over time and repeated builds, will the PSD of the feedstock powder change?

Comprehensive mechanical testing

- What is the impact on fatigue performance?
- How is tensile strength affected?
- Will the microhardness change?

Lab work

- How will the chemistry (oxygen uptake) of the feedstock change over time?
- How will the PSD change over time?

Improvements

- Can the surface quality be improved upon?

2. Thicker layers: Density of specimens built in 120 μm layers.

Building on previous work:

- a. 7 mm defocus (below the focal plane)**
- b. 8 mm defocus (below the focal plane) in**

Standard powder (15-53 μm) in Thicker layers (120 μm)

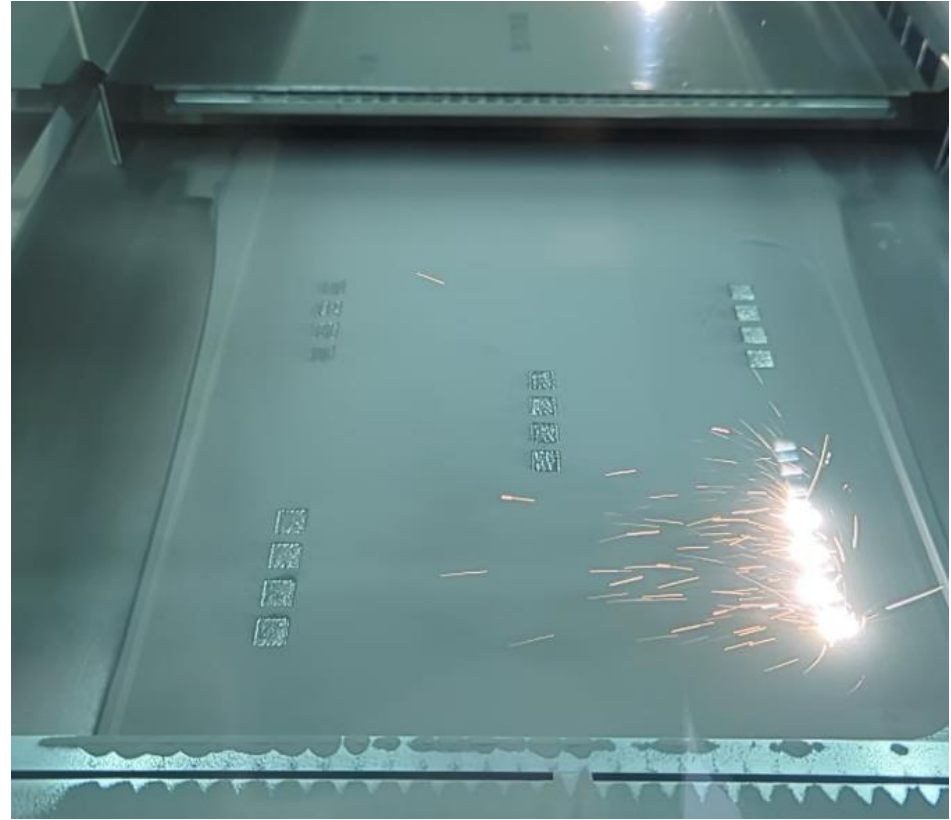
2. Standard powder (15-53 μm PSD) in Thicker layers (120 μm)

	60 μm layer thickness	120 μm layer thickness
VED (J/mm^3)	37	26
Power (W)	320	448
Hatch (mm)	0.095	0.169
Scan Speed (mm/s)	1500	850
Layer thickness (mm)	0.06	0.12
Beam diameter (mm)	0.080	$\approx 0.161 - 0.179$

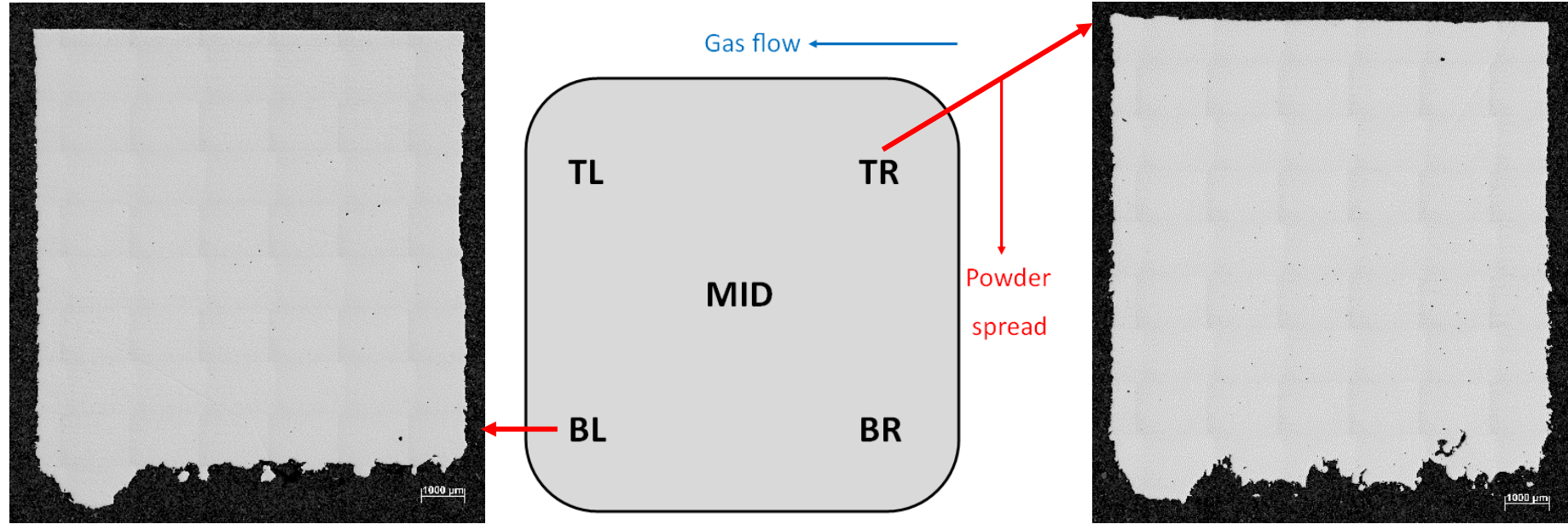
- Following on from previous work that developed laser parameters for producing parts in 120 μm layer thickness, this build was carried out to assess the affect of build location on the density of the samples.
- By dropping the focus of the laser to below the surface of the build, it widens the spot-size of the laser at the site of the build plane. This will have the effect of changing the shape of the melt pool with the intention being to reduce the formation of keyhole pores, to suppress the generation of spatters and to limit the effects of vaporisation, such as the attenuation of the laser beam.
- The laser is calibrated by the machine manufacturer during servicing of the machine and the beam diameters were provided.

2. Standard powder (15-53 μm PSD) in Thicker layers (120 μm)

- During the build, a great deal of spatter was produced and thrown across the surface of the powder bed: the laser power is high at 448 W in order to penetrate through the thicker layers of powder but may be creating an increase in the turbulence in the meltpool.
- The types of spatter produced included the hot particles seen in these images, as well as a large amount of finer black particulates.

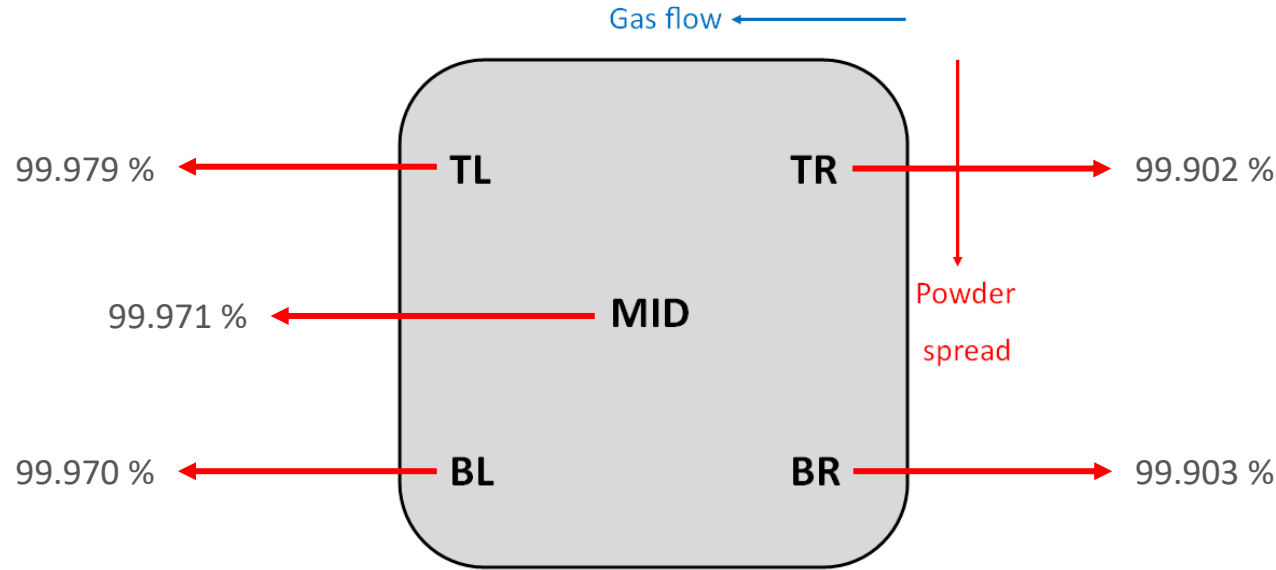


2a. Standard powder (15-53 μm PSD) in Thicker layers (120 μm) – 7 mm



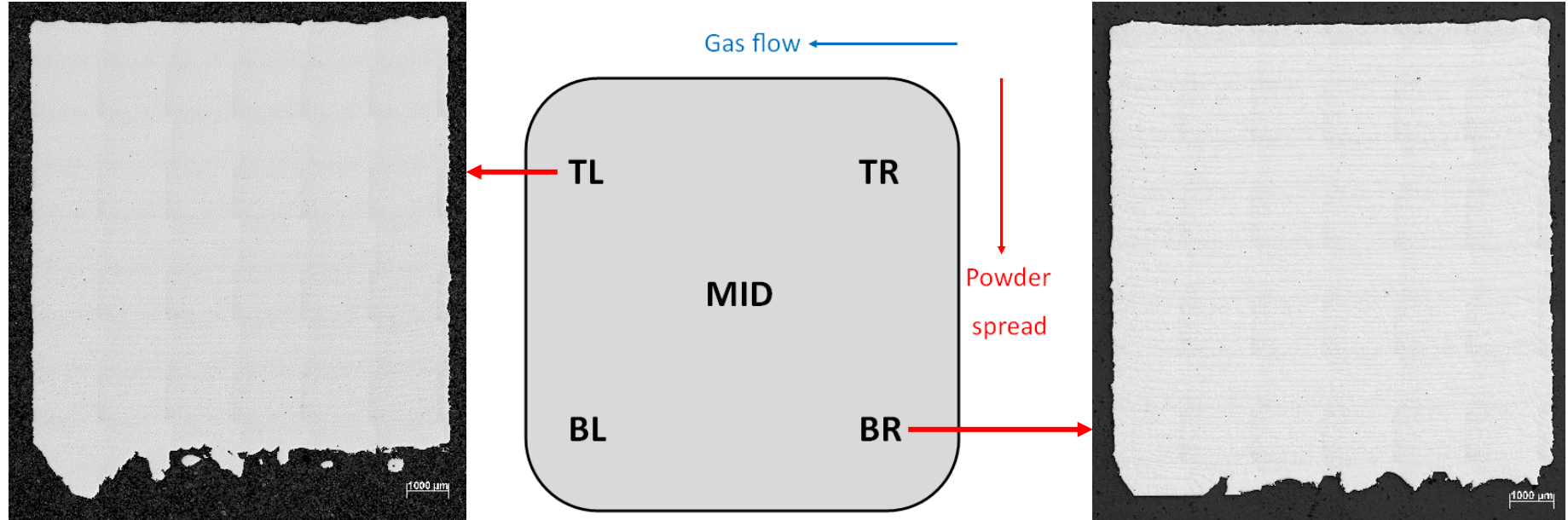
Four (4) cube samples from each build location were sectioned in half using the Secotom, a desktop precision sectioning machine in the build (Z) direction. The samples were then hot mounted in edge-retentive Bakelite. The samples were ground with SiC paper (#220 grit), then polished with diamond grit (9 μm) and finished with 0.04 μm colloidal silica. For the assessment of the density, the samples were imaged using the Zeiss Axio microscope with the porosity (density) being measured using the built-in software.

2a. Standard powder (15-53 μm PSD) in Thicker layers (120 μm) – 7 mm



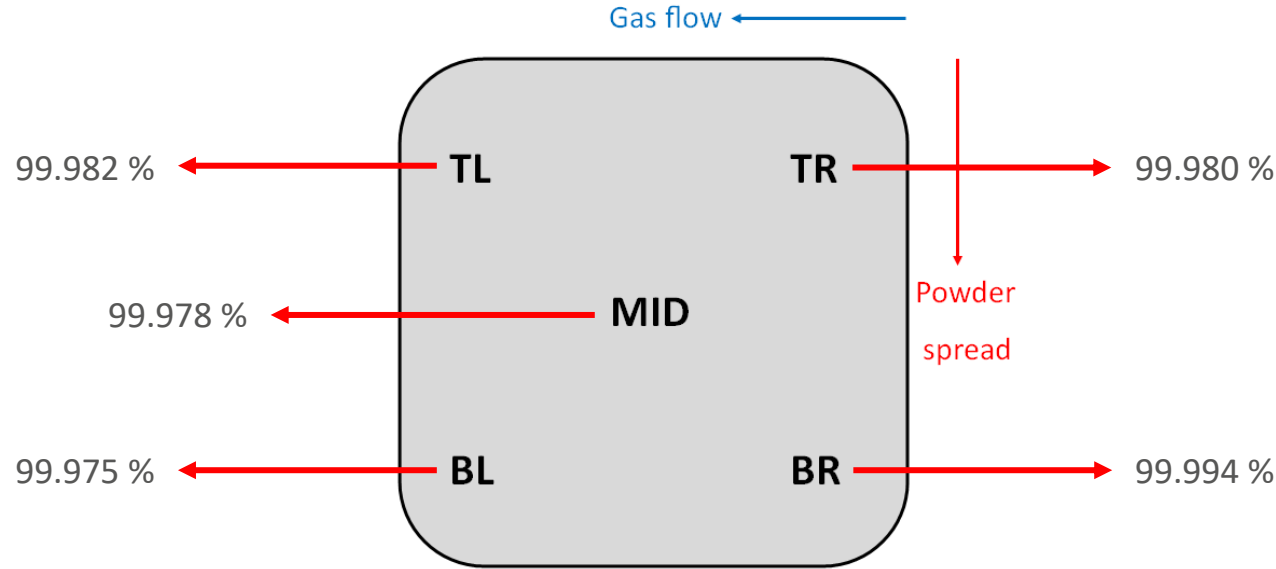
Four (4) cube samples from each build location were measured for porosity with the figures above showing the average density for the given build location. Standard deviation was calculated using the Excel Standard Deviation function.

2b. Standard powder (15-53 μm PSD) in Thicker layers (120 μm) – 8 mm



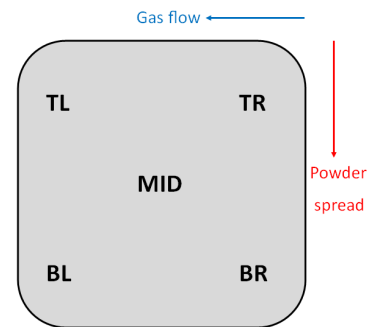
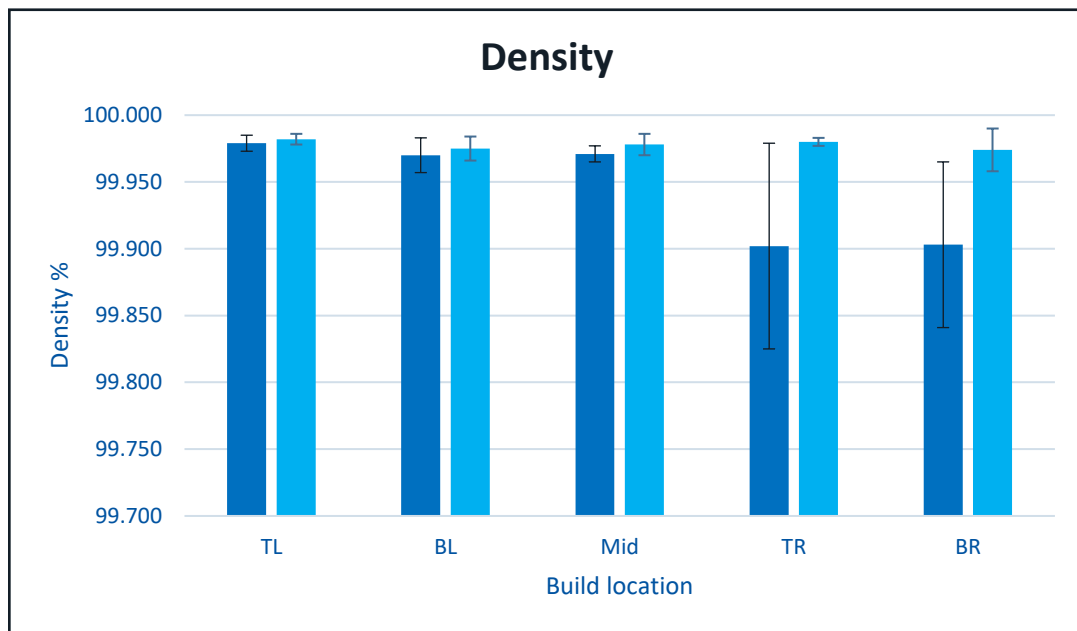
The defocus of 7mm and 8mm was assigned to cubes and built stacked on top of each other.

2a. Standard powder (15-53 μm PSD) in Thicker layers (120 μm) – 8 mm



Four (4) cube samples from each build location were measured for porosity with the figures above showing the average density for the given build location. Standard deviation was calculated using the Excel Standard Deviation function.

2. Standard powder (15-53 μm PSD) in Thicker layers (120 μm)



- Overall, the density obtained was very high. Some of the lower measurements of density of around 99.8 % have resulted in larger error bars on the 7 mm specimens.
- For this machine, it was found that the 8 mm defocus achieved the highest and most consistent density results across the build platform, with density measured at between 99.950 – 99.986 %.

2. Thicker layers (120 μm) builds – Observations and conclusions

- Thicker layers need increased laser power therefore produce a less stable meltpool – which produces more spatter.
- Where the spatter falls can effect the chemical properties of the solid metal parts produced in LPBF-AM, if the spatter falls back on the bed and remelted by the laser.
- Increased spatter can reduce the 'length of the life' of the feedstock powder if it moves out of spec for oxygen content.
- The effect of build location of the mechanical performance of the samples – what effect does the remelting of spatter into samples/parts have on mechanical performance? Renishaw have reported lower tensile strength results with these set of parameters for building in 120 μm layers.
- Concerns fed back from Renishaw about the ability of the machine to change rapidly with the changing defocus.

Advantages of developing 90 μm layer thickness parameters

- Reduction of power may improve melt pool stability and produce less spatter.
- Decrease in the impact of the 'stair-stepping' effect to improve on surface quality.

3. Improving Surface Quality: Developing parameters for 90 μm layers.

Standard powder (15-53 μm) in Thicker layers (90 μm)

3. Developing parameters for 90 µm layers in Standard powder (15-53 µm)

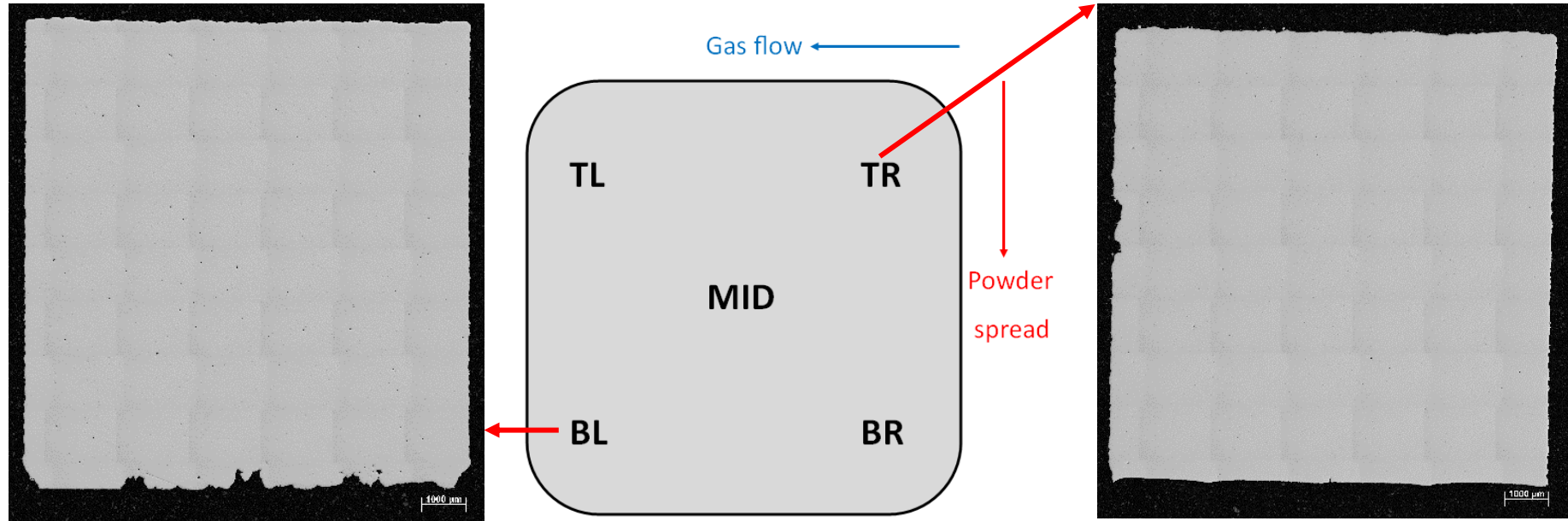
- The next step is to decrease the layer thickness, achieve high density in the specimens produced whilst maintaining

	60 µm layer thickness	90µm layer thickness	120 µm layer thickness
VED (J/mm³)	37	24	26
Power (W)	320	425	448
Hatch (mm)	0.095	0.140	0.169
Scan Speed (mm/s)	1500	1400	850
Layer thickness (mm)	0.06	0.09	0.12
Beam diameter (mm)	0.080	0.080	≈ 0.161 – 0.179

the same production rate achieved in the manufacture of the 120 µm samples.

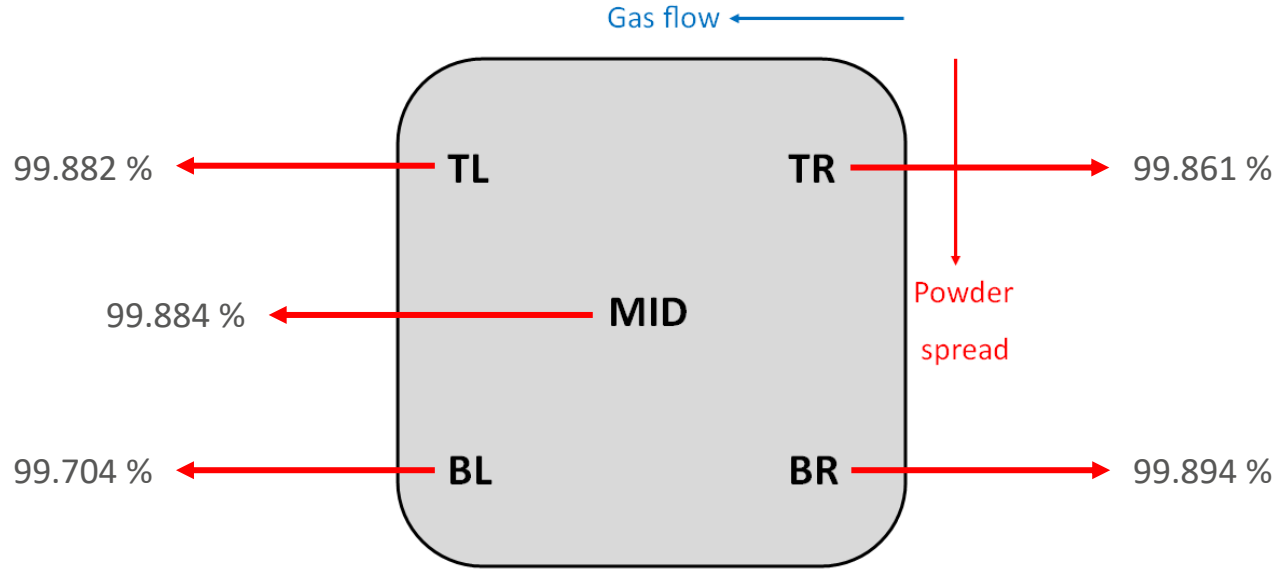
- This set of laser parameters was developed in conjunction with engineers at Renishaw, who carried out some initial cube builds in order to narrow down parameter that could build 90 µm layer specimens with high density in standard powder, whilst maintaining the same build rate achieved with the 120 µm layer thickness.
- For this, the power and the hatch distance was reduced and the scan speed was increased. The spot size was returned to 0.08 mm, the same as with the 60 µm layer thickness – no defocus.

3. Standard powder (15-53 μm PSD) in Thicker layers (90 μm)



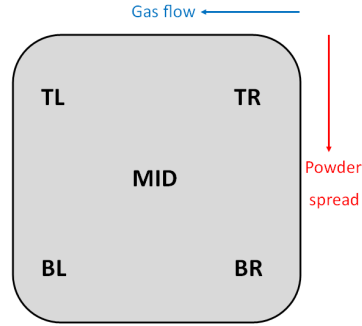
Four (4) cube samples from each build location were sectioned in half using the Secotom, a desktop precision sectioning machine in the build (Z) direction. The samples were then hot mounted in edge-retentive Bakelite. The samples were ground with SiC paper (#220 grit), then polished with diamond grit (9 μm) and finished with 0.04 μm colloidal silica. For the assessment of the density, the samples were imaged using the Zeiss Axio microscope with the porosity (density) being measured using the built-in software.

3. Standard powder (15-53 μm PSD) in Thicker layers (90 μm)

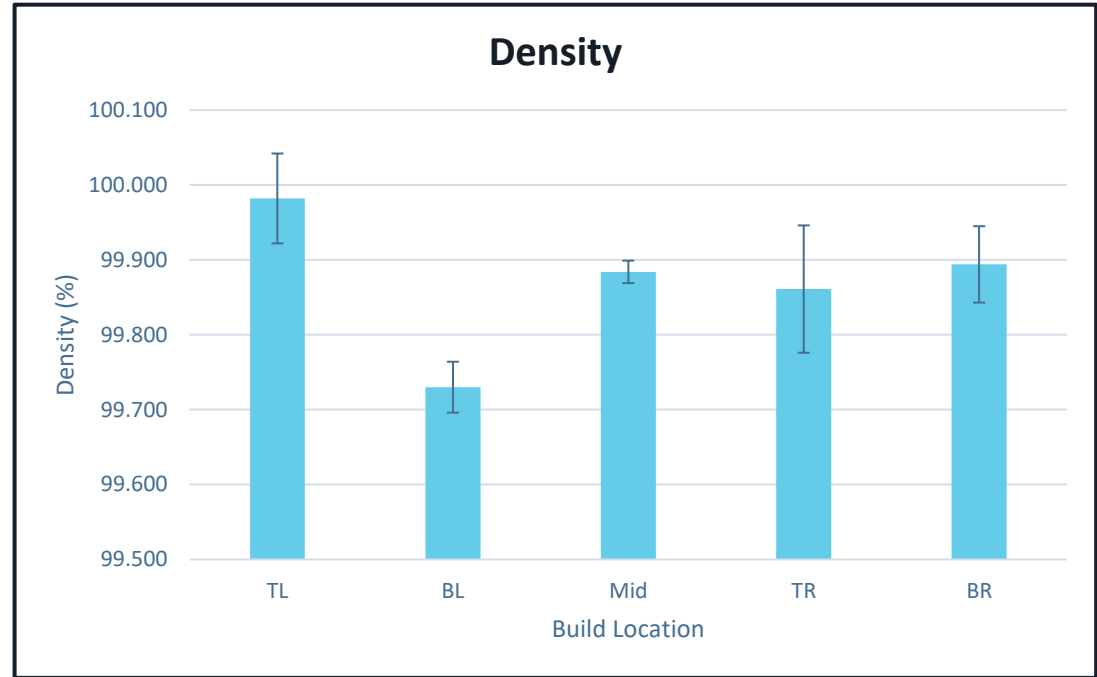


Four (4) cube samples from each build location were measured for porosity with the figures above showing the average density for the given build location.

3. Standard powder (15-53 μm PSD) in Thicker layers (90 μm)



- Standard deviation was calculated using the Excel Standard Deviation function and applied to give the error bars.
- Overall, the density obtained was lower than that achieved in the 120 μm layer specimens. The density across the build platform ranged between 99.704 - 99.958 %.



- The cubes built in the bottom-left area of the had a lower density than those specimens built in the other platform locations.

3. Standard powder (15-53 μm PSD) in Thicker layers (90 μm) – observations and conclusions.

- Density across the build plate varied between 99.704 % to 99.958 %, with the bottom-left area showing the lowest density of 99.704 % on average. This is the area of the build furthest away from the powder spread and the gas flow outlet. It is also the side of the bed on which the most spatter is expected to fall, although the density was the highest on average in the top left area of the build platform – an area also expected to receive higher quantities of spatter.
- An examination of the gas flow across the platform will be carried out to see if there are any areas that experience gas flow variations.

4. Building in Coarser powder with a comparison of Standard and Thicker Layers:

4a. Density

4b. PSD changes (over time)

4c. Oxygen uptake (over time)

4d. Surface roughness

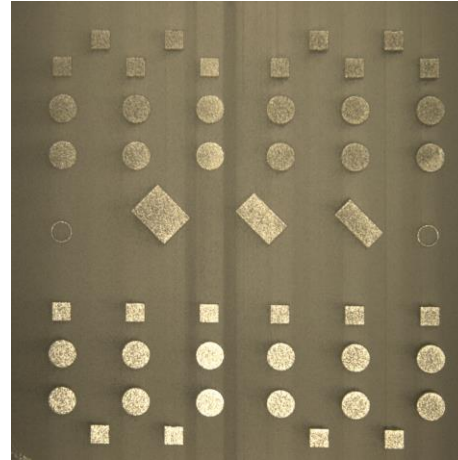
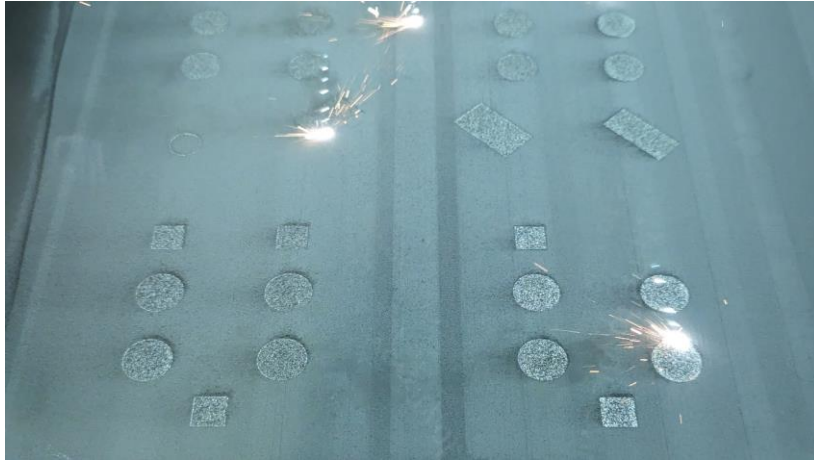
4e. Microhardness

4f. Tensile properties

4g. Fatigue properties

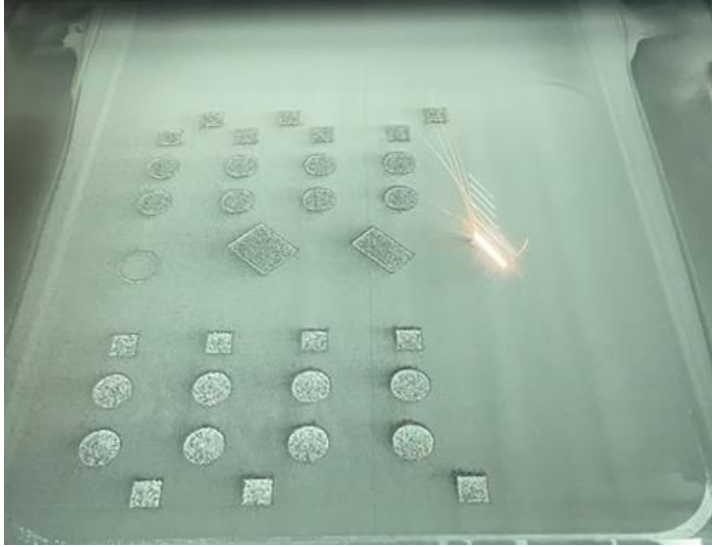
4h. Gas flow properties

4. Building in Coarser powder (15-106 μm) in Standard layers (60 μm)



- Furrows were observed in powder bed – images taken from the *in situ* camera appear to show this is due to the layout of the bars on the build platform.
- Following the completion of the cube specimens, the furrows in the powder bed disappeared.
- An overlapping or staggered layout of the bars should eliminate this phenomena.

4. Building in Coarser powder (15-106 μm) in Thicker layers (90 μm)



- The 90 μm layer thickness build appeared not to have the same furrowing effect observed in the 90 μm layer thickness build.



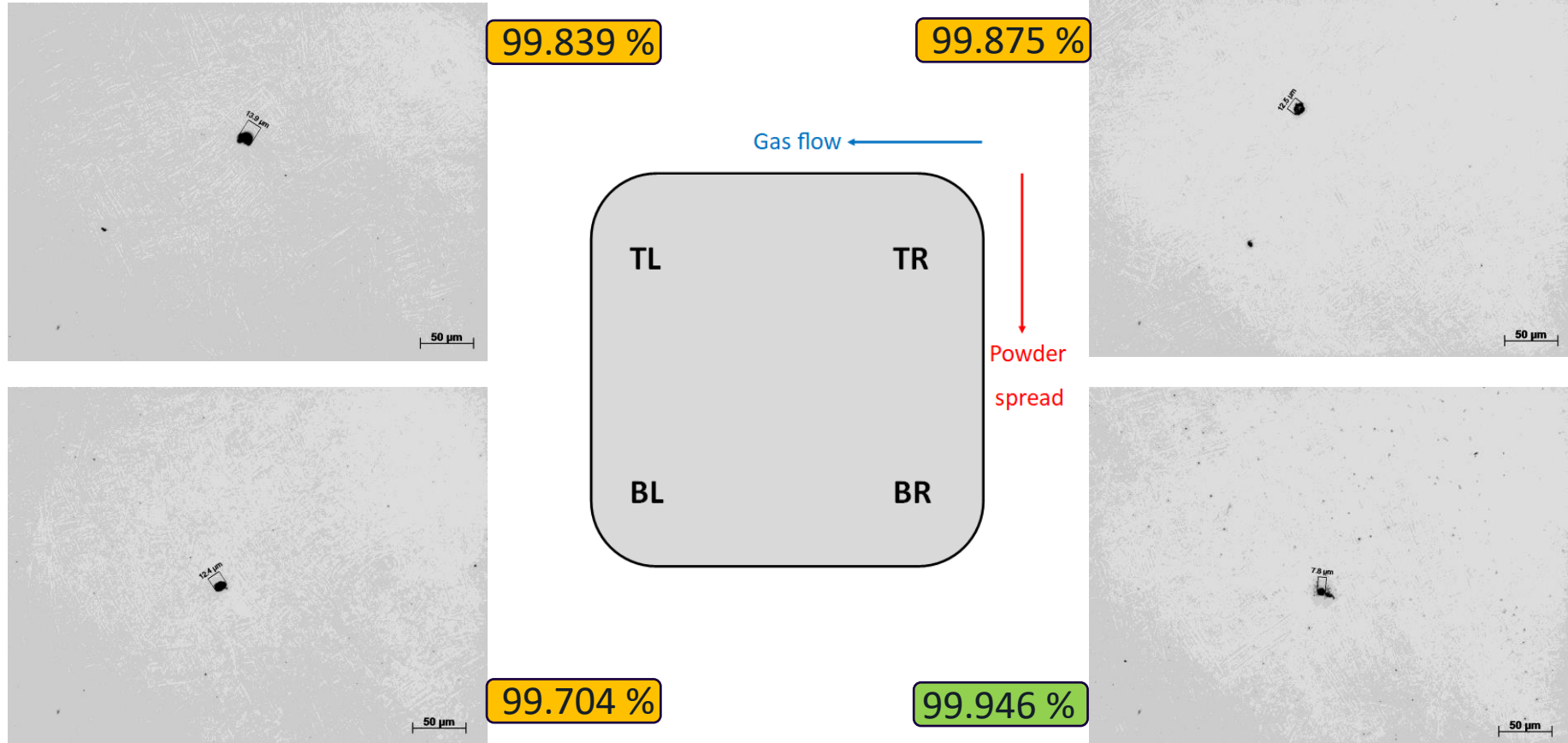
- The build with a 60 μm layer thickness was set up with line supports, however, whilst the build did complete successfully, the adhesion of the specimens to the substrate was found to be weak – the specimens were able to be removed easily by hand. For this 90 μm layer thickness build, pyramid supports were employed to achieve a better contact and reduce the chance of a failed build.

4a. Density: a comparison

- To examine the effect of building in a coarser powder (15-106 μm) on the porosity, density measurements were carried out on cube specimens taken from the four quadrants of the build platform.
- One cube sample from each build location were sectioned in the as built (Z direction) using the Secotom, a desktop precision sectioning machine.. The samples were then hot mounted in edge-retentive Bakelite. The samples were ground with SiC paper (#220 grit), then polished with diamond grit (9 μm) and finished with 0.04 μm colloidal silica. For the assessment of the density, the samples were imaged using the Zeiss Axio microscope with the porosity (density) being measured using the built-in software.
- Three build conditions were examined:
 - i. **Standard powder (15-45 μm) in Thicker layers (90 μm)** (results taken from the build outlined in section 3 of this presentation)
 - ii. **Coarser powder (15-106 μm) in Standard layers (60 μm)**
 - iii. **Coarser powder (15-106 μm) in Thicker layers (90 μm)**
- The following three slides show higher magnification (x200) images and a measurement of the largest pore or defect observed in that image as well as the measured density for that build location.

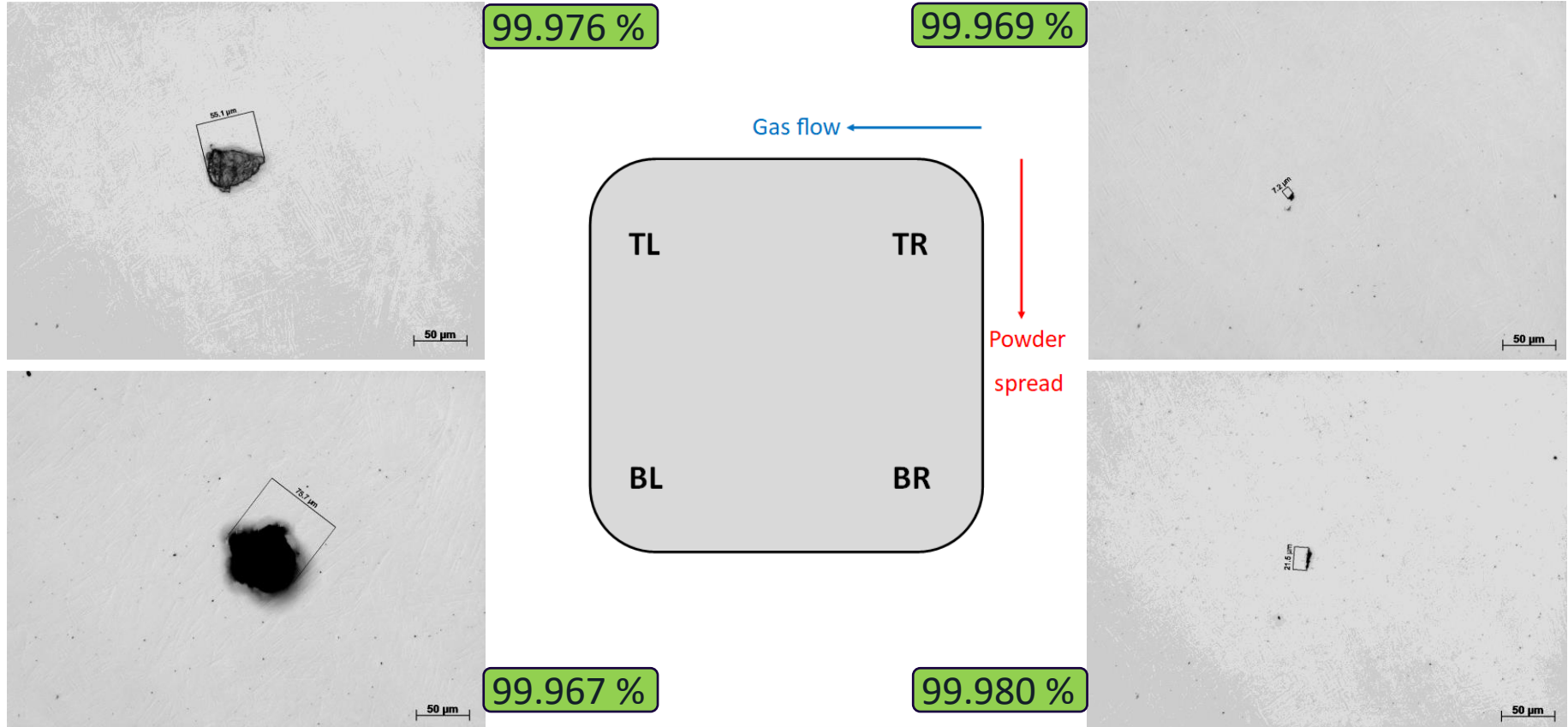
4a. Density

i. Standard powder (15-45 μm) in Thicker layers (90 μm)



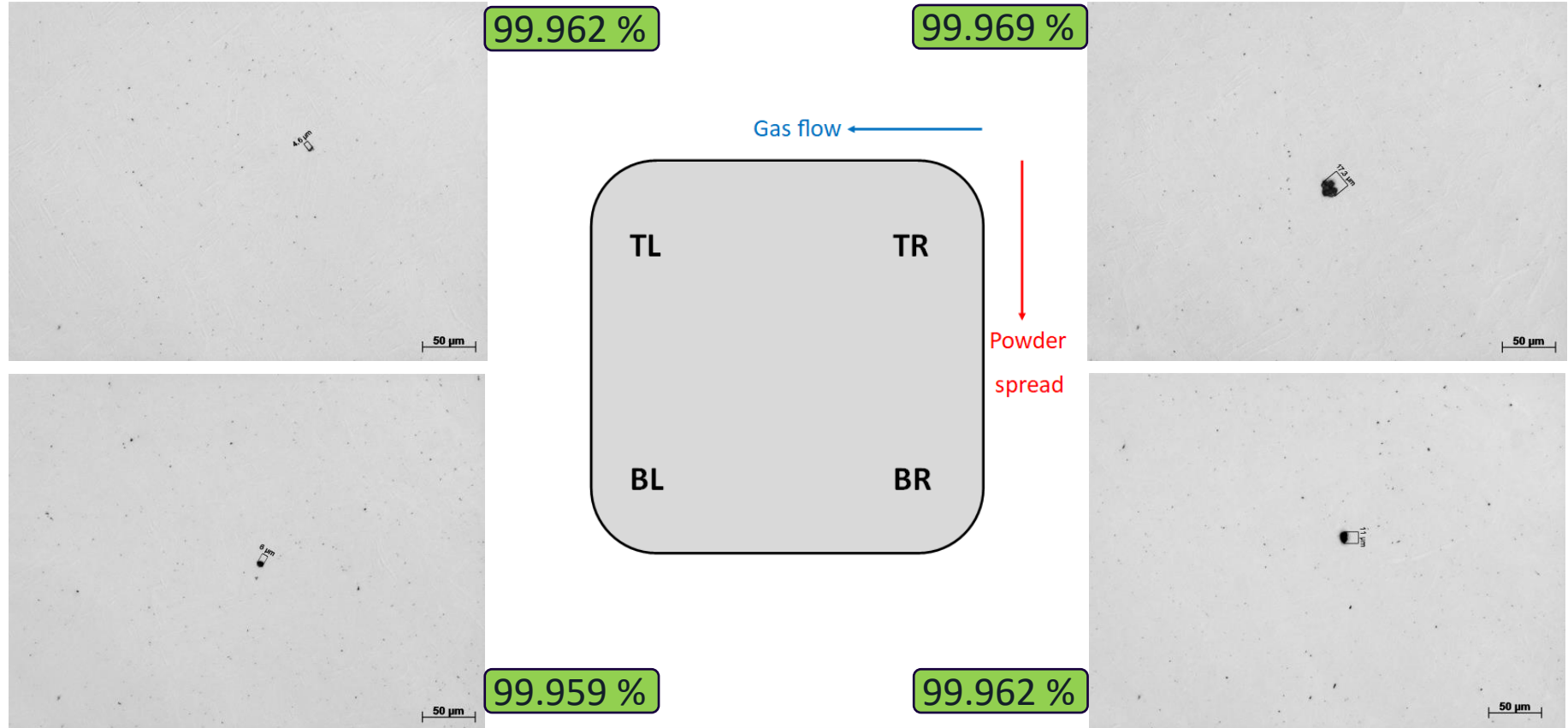
4a. Density

ii. Coarser powder (15-106 μm) in Standard layers (60 μm)



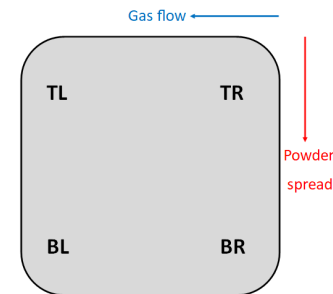
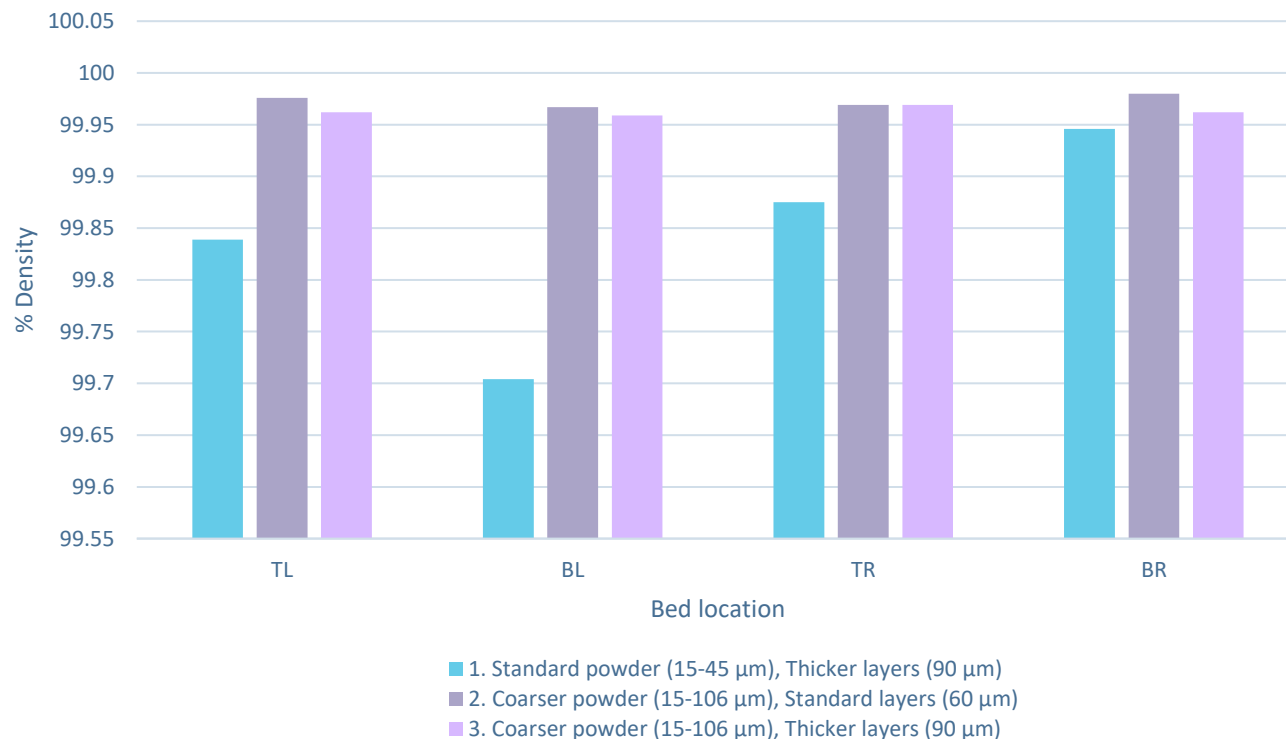
4a. Density

iii. Coarser powder (15-106 μm) in Thicker layers (90 μm)



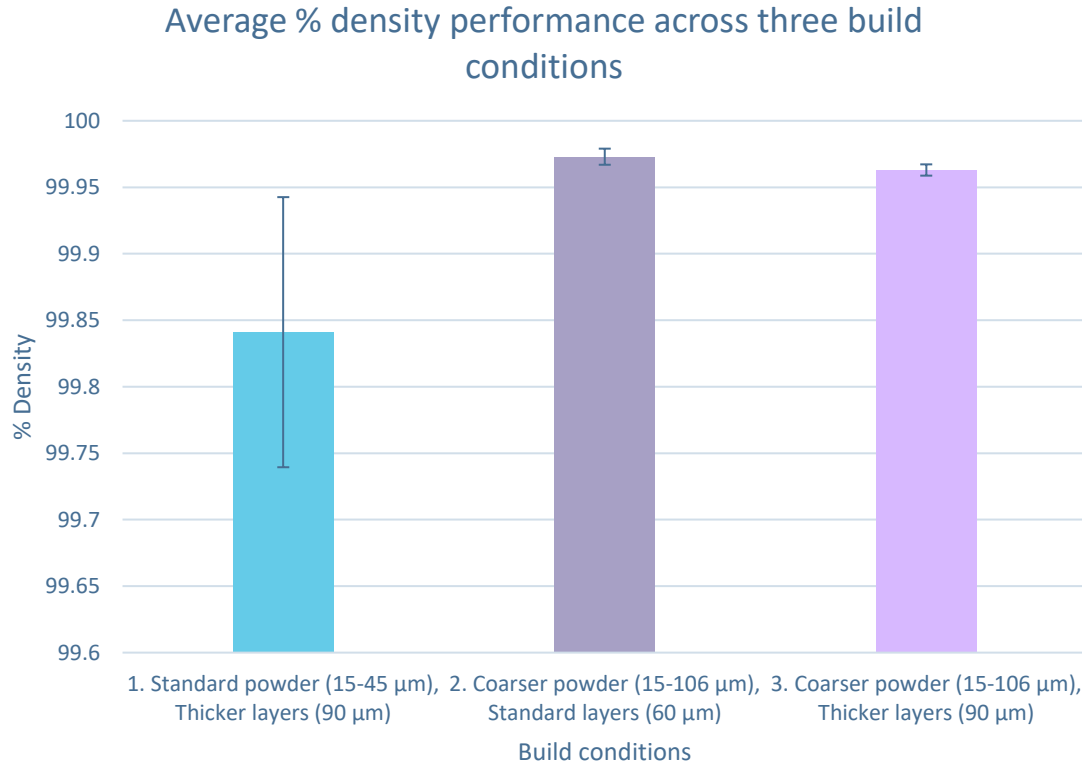
4a. Density: a comparison of all three build conditions

% Density performance across three build conditions



There are no error bars for these results as the density was measured in only one specimen from each build location.

4a. Density: a comparison



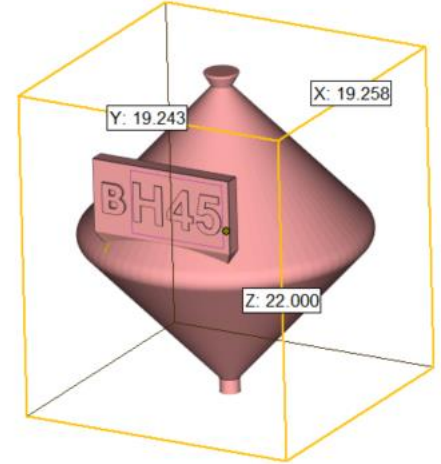
In this graph, an average of the density measured in the four build locations was calculated and plotted. There are four measurements from each build, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.

4a. Density – observations and conclusions

- Density is lower in the standard powder (99.841 %) when compared with the coarser powder (99.963 %) with the thicker layers (90 μm), although the difference is not large. The small difference may be due to differences in porosity in the powder bed between the two powders. The packing densities of the two powders will be examined.
- In the coarser powder, density is slightly better in the 60 μm layers (99.973 %) than in the 90 μm layers (99.963 %), although again, the difference is very small. In these two builds, the difference is the laser parameters, most importantly the power.
- Porosity observed in specimens built with the coarser powder can be characterised as small (sub 1 μm) but numerous whereas the porosity in specimens built with the standard powder appears to be fewer in the number of small pores, but an increased presence of larger pores (multiple pores of between 10 and 20 μm observed).

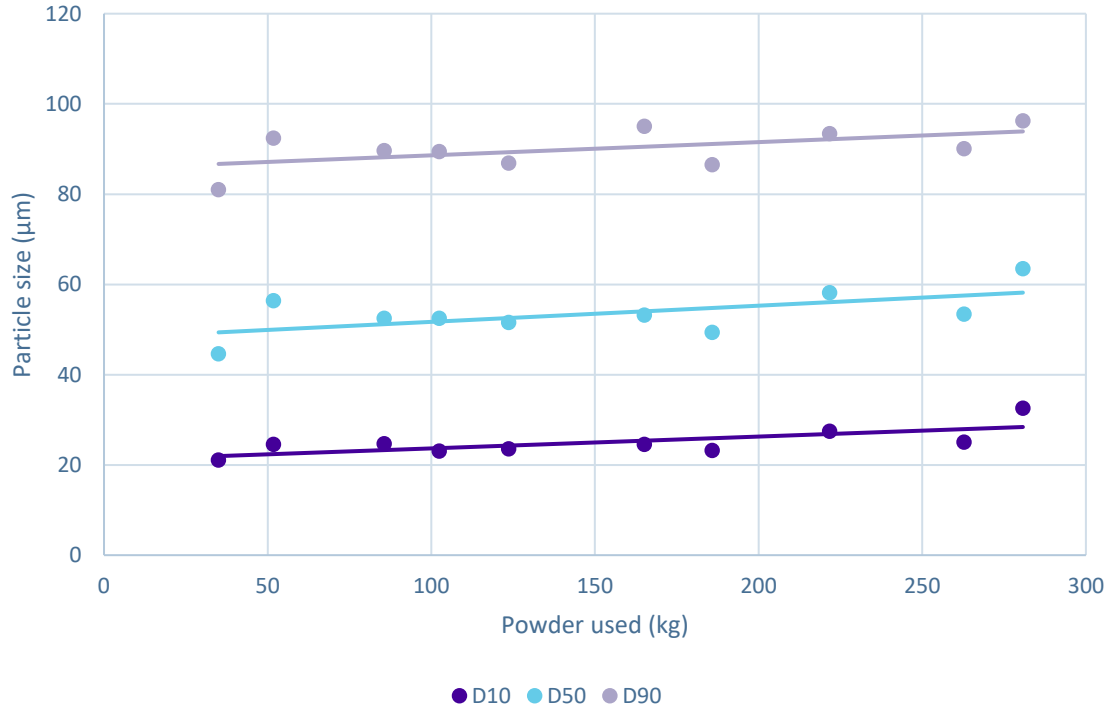
4b. PSD measurements - method

- To find out whether the PSD changes over the course of many builds, powder samples were taken from the bed for each build.
- The 10 builds that were carried out in the coarser range of powder were all different from each other: some were short cube, single laser builds, others were tall, multi-laser builds. Some were builds from other projects .
- The 10th build was a test of a novel shielding gas mixture (Ar/He) and was not optimised – it produced a great deal of spatter that fell onto the bed. The results are reported later in brackets.
- For the builds that were designed for this project, powder capsules were added to capture powder from the bed at the start, one on the left of the build & one on the right. This was added to the build layout to ensure the capture of a powder bed sample for analysis later & to avoid contamination.
- The PSD analysis was carried out on the Camsizer X2 and there were 2 repeats for each powder sample, with the results averaged.
- The PSD results were plotted against the estimated amount of powder needed for each build to give an indication as to how many times the powder has gone around the closed loop system.



4b. PSD measurements

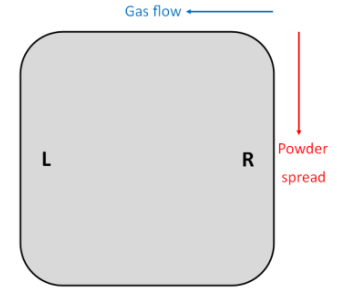
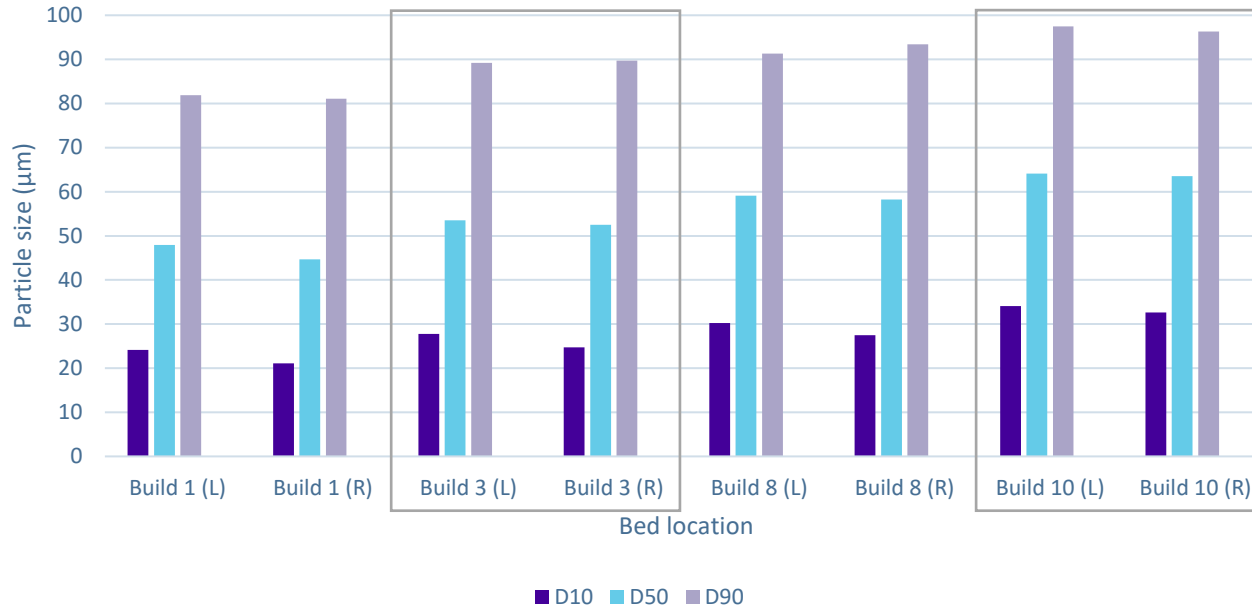
Change of particle size distribution over 10 builds



The PSD measurements are plotted against the amount of powder estimated to be required for each build. The machine was loaded with 90 kg of virgin 15-106 µm Ti-6Al-4V powder at the start of the 10 builds. It is estimated that the powder has recirculated 3-4 times throughout the course of these 10 builds, although this is difficult to determine, given the close-loop system of the in built sieve and cyclone of the Ren 500Q.

4b. PSD measurements

Comparing changes in particle size distribution depending on bed location



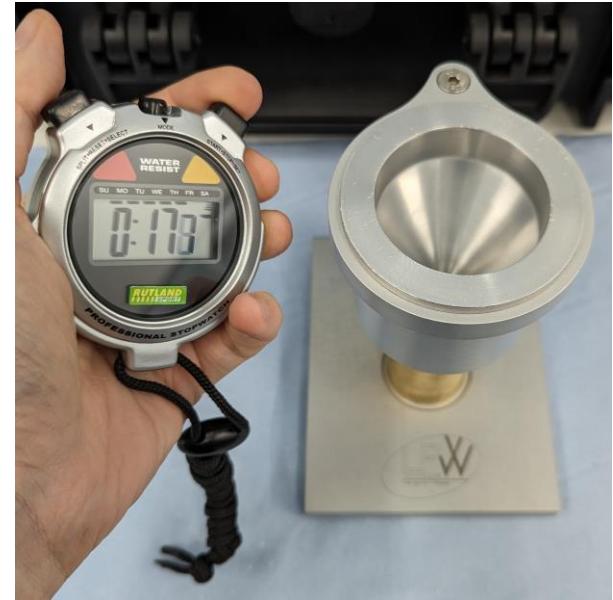
For the builds with powder capsules built on both sides of the bed, the PSD has been compared in this graph to determine whether the fall of spatter impacts on the PSD of the powder bed of the left compared with the right.

4b. PSD measurements – observations and conclusions

- The D10, D50 and the D90 have all increased over the course of the 9 (10) builds:
D10 = 21.1 μm $\rightarrow$ 25.1 μm (32.6 μm)
D50 = 44.7 μm $\rightarrow$ 53.5 μm (63.5 μm)
D90 = 81.1 μm $\rightarrow$ 90.1 μm (96.3 μm)
- It is expected that the PSD would have increased quantities of larger particles on the left side of the bed when compared to the right and this trend is seen in Build 1 and Build 10 (although this build produced very large quantities of spatter and the gas flow was not optimised for the shielding gas).
- However, there are varied results for the other builds in the D90 results:
Build 3 has larger particles on the right for the D90, although there is little difference, (L = 89.2 μm , R = 89.7 μm).
Build 8 has larger particles on the right for the D90, (L = 91.3 μm , R = 93.5 μm).
- Overall, whilst there are some changes in the PSD across the course of the ten (10) builds carried out in this coarser range of powder, the changes are minimal and further builds should be carried out to determine whether this shift continues.

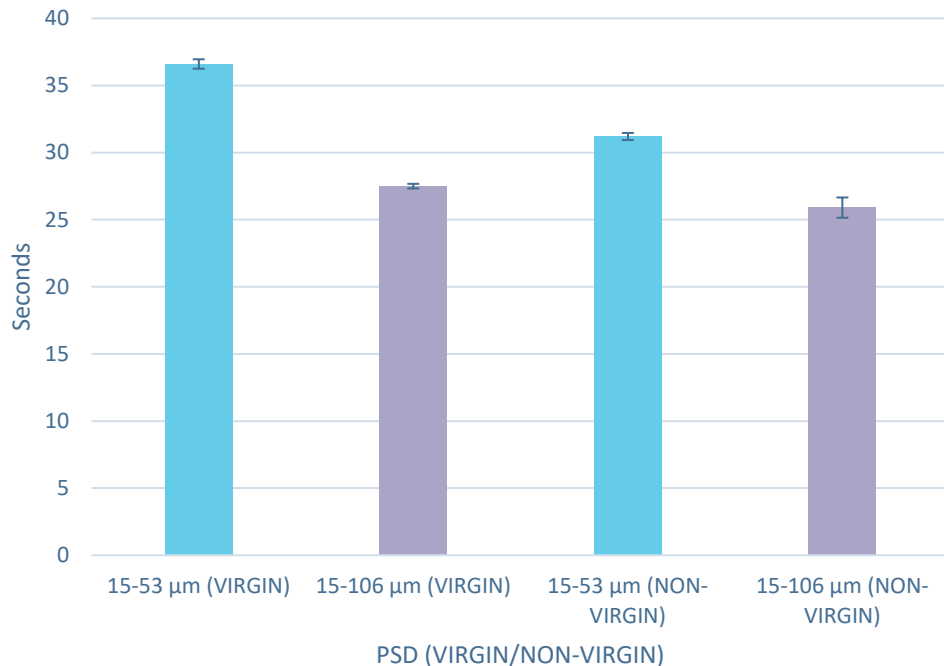
4b. Flowability measurements – method

- The flowability of the powder was measured using the Hallflow Static Flow Method 1 (Stationary Powder) according to the standard ASTM B213-13.
- Four (4) powders were examined and compared:
 - i. 15-53 μm VIRGIN
 - ii. 15-53 μm NON-VIRGIN (after more than 10 builds)
 - iii. 15-106 μm VIRGIN
 - iv. 15-106 μm NON-VIRGIN (after 10 builds)
- The measurements were carried out in a temperature and humidity controlled environment, with three repeats. Humidity was recorded as 37.0 % and the temperature at 22 °C.
- The three repeated measurements were recorded, averaged and with the correction factor applied, the results are shown on the next slide.



4b. Flowability measurements – results

A comparison of flowability between the Standard powder (15-53 μm) and the Coarser powder (15-106 μm) in a virgin and non-virgin state

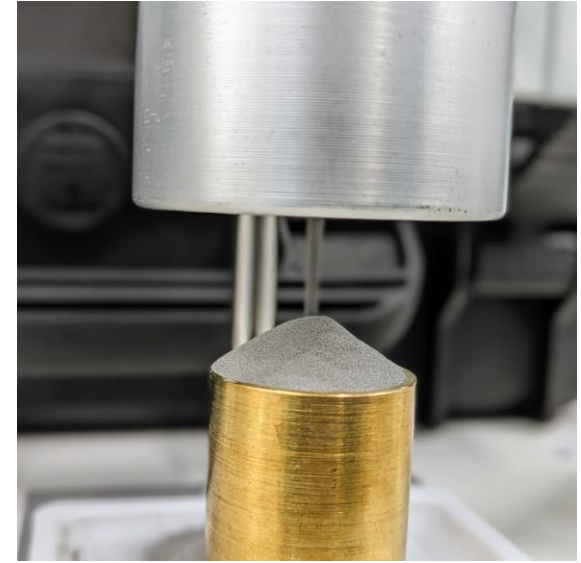


Powder PSD and status	Hallflow (s)
15-53 μm VIRGIN	36.6
15-53 μm NON-VIRGIN	31.2
15-106 μm VIRGIN	27.5
15-106 μm NON-VIRGIN	25.9

- Flowability is improved with use in both powder size ranges.
- Flowability is better with the coarser powder (15-106 μm) than with the standard powder (15-53 μm) in both the virgin and the used state.

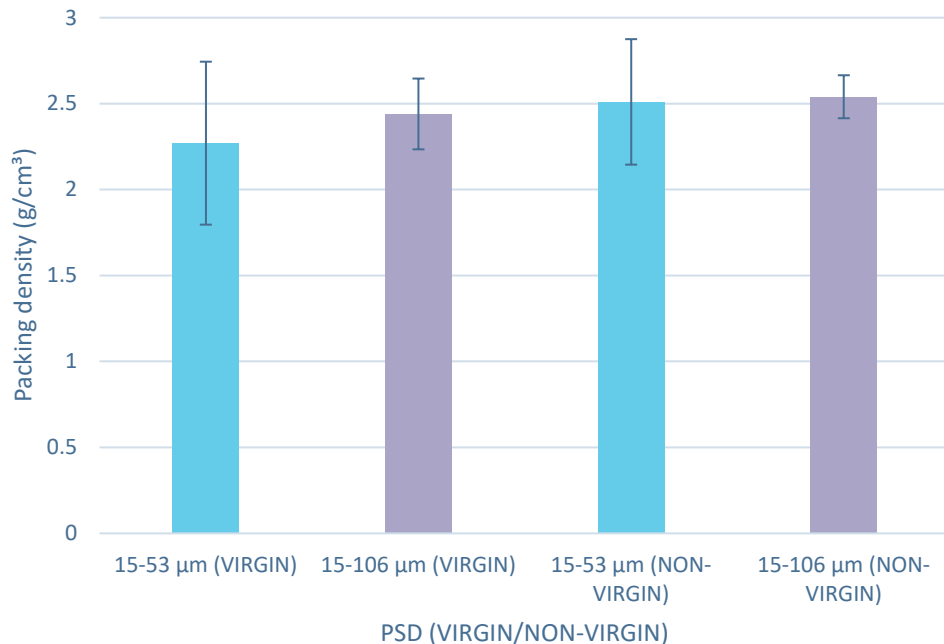
4b. Packing density measurements – method

- The packing apparent density of the powders was measured using the Apparent Density of Free-Flowing Metal Powders method, according to the standard ASTM B212-13.
- Four (4) powders were examined and compared:
 - i. 15-53 μm VIRGIN
 - ii. 15-53 μm NON-VIRGIN (after more than 10 builds)
 - iii. 15-106 μm VIRGIN
 - iv. 15-106 μm NON-VIRGIN (after 10 builds)
- The measurements were carried out in a temperature and humidity controlled environment, with three repeats. Humidity was recorded as 37.0 % and the temperature at 22 °C.
- The three repeated measurements were recorded, averaged and with the correction factor applied, the results are shown on the next slide, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.



4b. Packing density measurements – results

A comparison of the packing density between the Standard powder (15-53 μm) and the Coarser powder (15-106 μm) in a virgin and non-virgin state



Powder PSD and status	Packing density (g/cm ³)
15-53 μm VIRGIN	2.27
15-53 μm NON-VIRGIN	2.51
15-106 μm VIRGIN	2.44
15-106 μm NON-VIRGIN	2.54

- The packing density is higher with use in both powder size ranges.
- Packing density is higher with the coarser powder (15-106 μm) than with the standard powder (15-53 μm) in both the virgin and the used state.

4b. Flowability and packing density – observations and conclusions

- The flowability is improved and the packing density is greater after use in both powder size ranges.
- Flowability is better with the coarser powder (15-106 μm) than with the standard powder (15-53 μm) in both the virgin and the used state. There have been no issues with the powder flowing or using the coarser powder the integrated sieve system of Ren500Q.
- Packing density is higher with the coarser powder (15-106 μm) than with the standard powder (15-53 μm) in both the virgin and the used state. This could result in there being a greater quantity of powder being dosed and spread with each layer, however, there has been no issues with the machine coping with the coarser powder.

Further work:

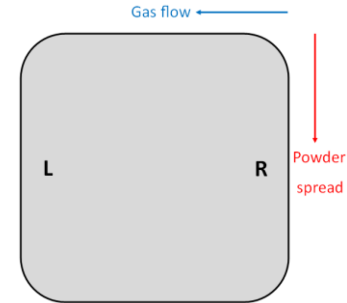
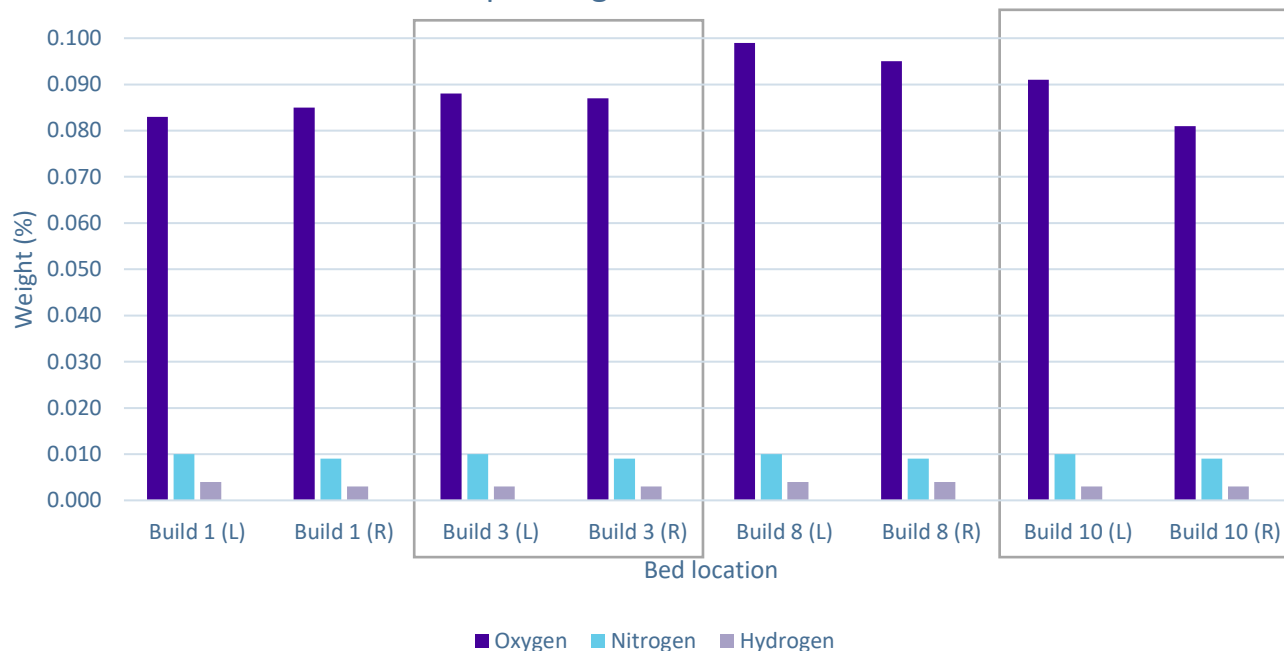
- A closer examination on the effects of an increased flowability and greater packing density, looking specifically at the affect on the thermal gradients, cooling rates and the resulting microstructure.

4c. Oxygen uptake (over time) - method

- To find out whether the chemistry (oxygen uptake) of the powder changes over the course of many builds, powder samples were taken from the bed for each build.
- The 10 builds that were carried out in the coarser range of powder were all different from each other: some were short cube, single laser builds, others were tall, multi-laser builds. Some were builds from other projects .
- The tests for 14 samples taken from 10 builds were carried out on a LECO 736 Series. The concentration of an unknown sample is determined relative to calibration standards.
- The following slides show firstly the results comparing any differences in the chemistry between the left side of the build platform and the right side, to determine whether the fall of spatter back down onto the bed could potentially cause a difference in the chemistry of the parts built in this location.
- Secondly, a look at the trend of the chemical changes of the powder over the course of the initial ten (10) builds is plotted and presented.

4c. Oxygen uptake (over time)

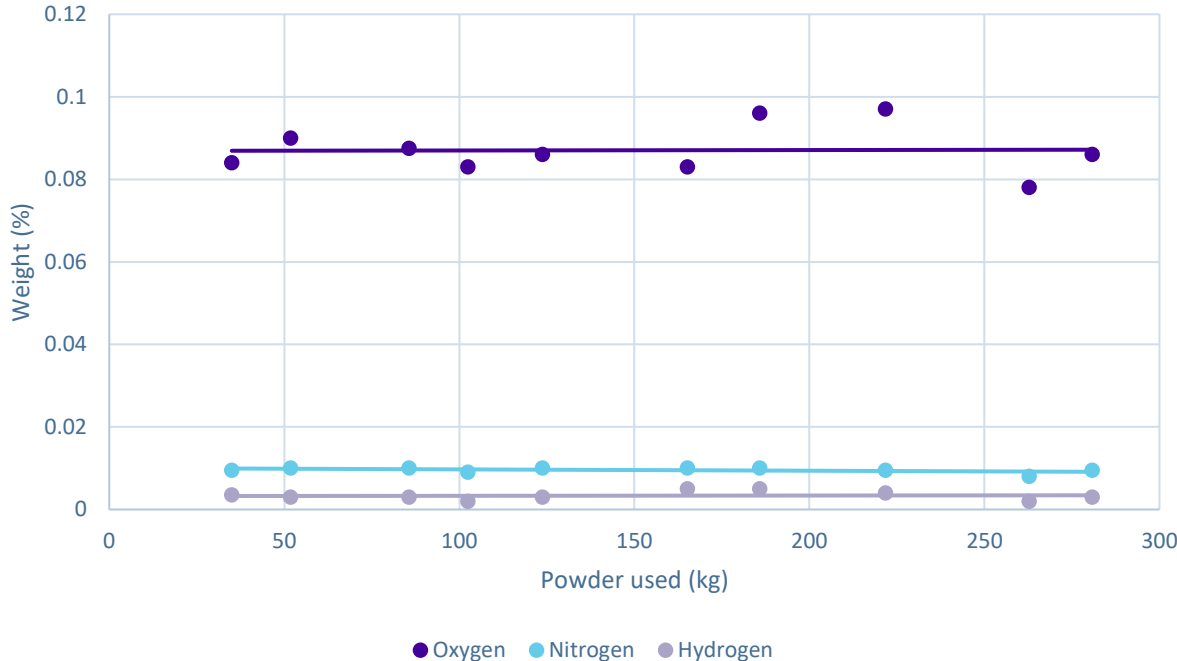
Comparing changes in oxygen, nitrogen and hydrogen quantites depending on bed location



For the builds with powder capsules built on both sides of the bed, the PSD has been compared in this graph to determine whether the fall of spatter impacts the powder bed chemistry of the left location compared with the right.

4c. Oxygen uptake (over time)

Change in oxygen, nitrogen and hydrogen over 10 builds



Summary

From measurements:

- Build 1 oxygen = 0.084 %
- Build 10 oxygen = 0.086 %

From the data sheet, for reference:

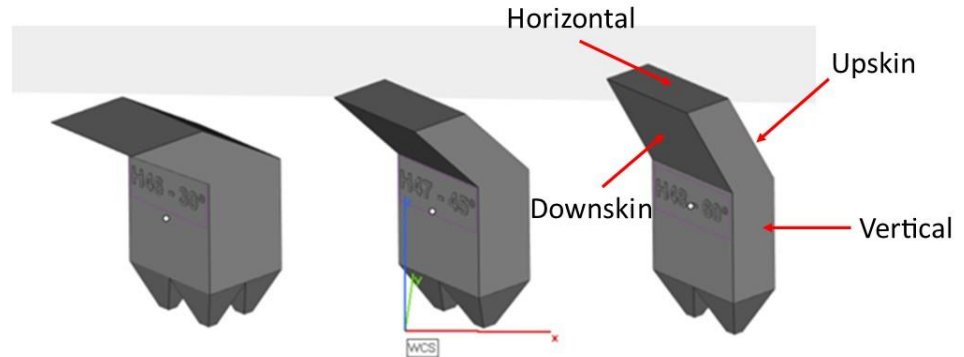
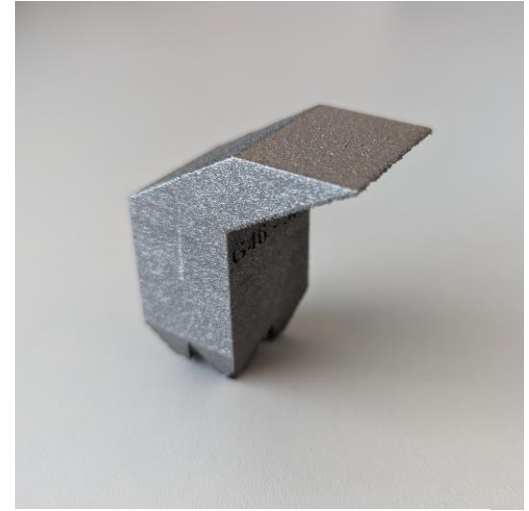
- Virgin oxygen content = 0.07 %
- Out of spec = 0.13 %

4c. Chemistry measurements – observations and conclusions

- Apart from Build 1, all samples taken from the left side of the build had a higher oxygen content than samples taken from the right (nearest the gas outlet), however, no significant differences measured.
- Over the course of the 10 builds, (an estimated 3 reuse rotations), there is no significant increase in the content of oxygen, (Build 1 = 0.084 % and build 10 = 0.086 %).
- Overall, whilst there are some changes in the chemistry across the course of the ten (10) builds carried out in this coarser range of powder, the changes are very minimal and further builds should be carried out to determine whether this shift continues.
- The powder has not be replenished over the course of the ten (10) builds, with virgin or blended sieved powder. It is likely the case that ten (10) builds are too few to see any trend or change in the chemistry and further study is needed.

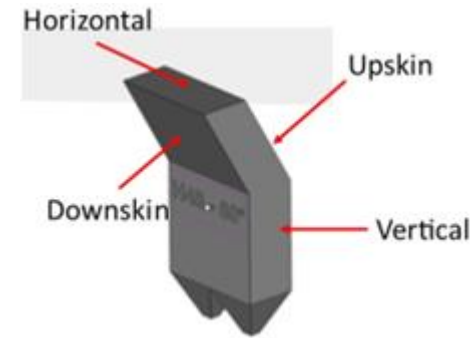
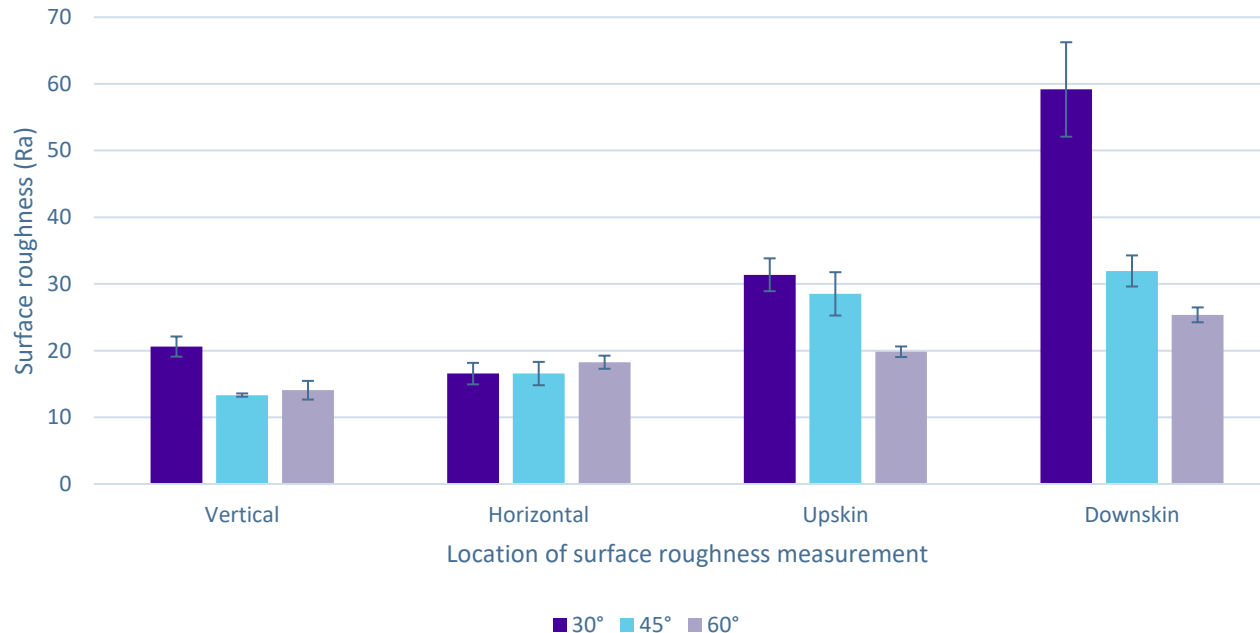
4d. Surface roughness - method

- In order to assess the affect of the coarser powder on the surface roughness in different layer thicknesses, blocks were designed to be able to measure 4 different surface on each specimen: the upskin, the downskin, the vertical and the horizontal surfaces.
- The blocks were printed with three different overhang angles: 60° , 45° and 30° .
- The measurements were carried out on a Keyence microscope due to the high surface roughness of the specimens.
- Five random locations on each of the four surfaces were imaged and measured and averaged, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.



4d. Surface roughness – 60 μm layer thickness

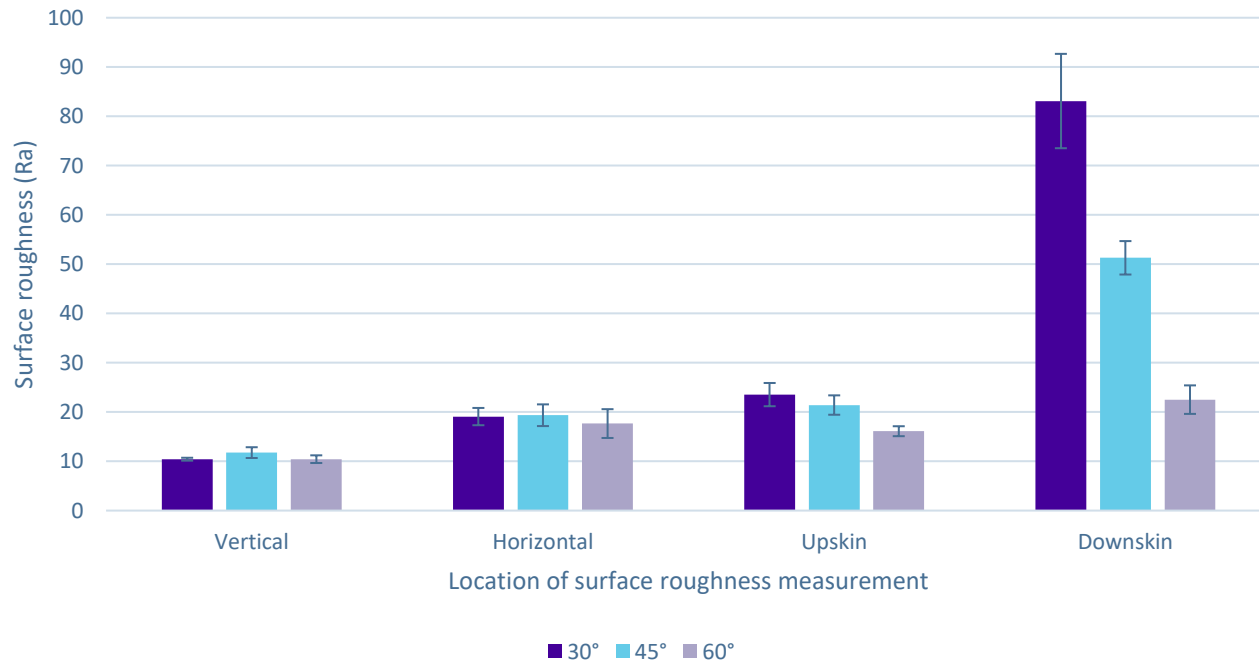
A comparison of the surface roughness on different surfaces and overhang angles in specimens built in 60 μm layers



- There is little difference seen in the vertical, horizontal and upskin surface roughness measurements.
- The surface roughness is highest on the downskin with the overhang angle of 30°.

4d. Surface roughness – 90 μm layer thickness

A comparison of the surface roughness on different surfaces and overhang angles in specimens built in 90 μm layers

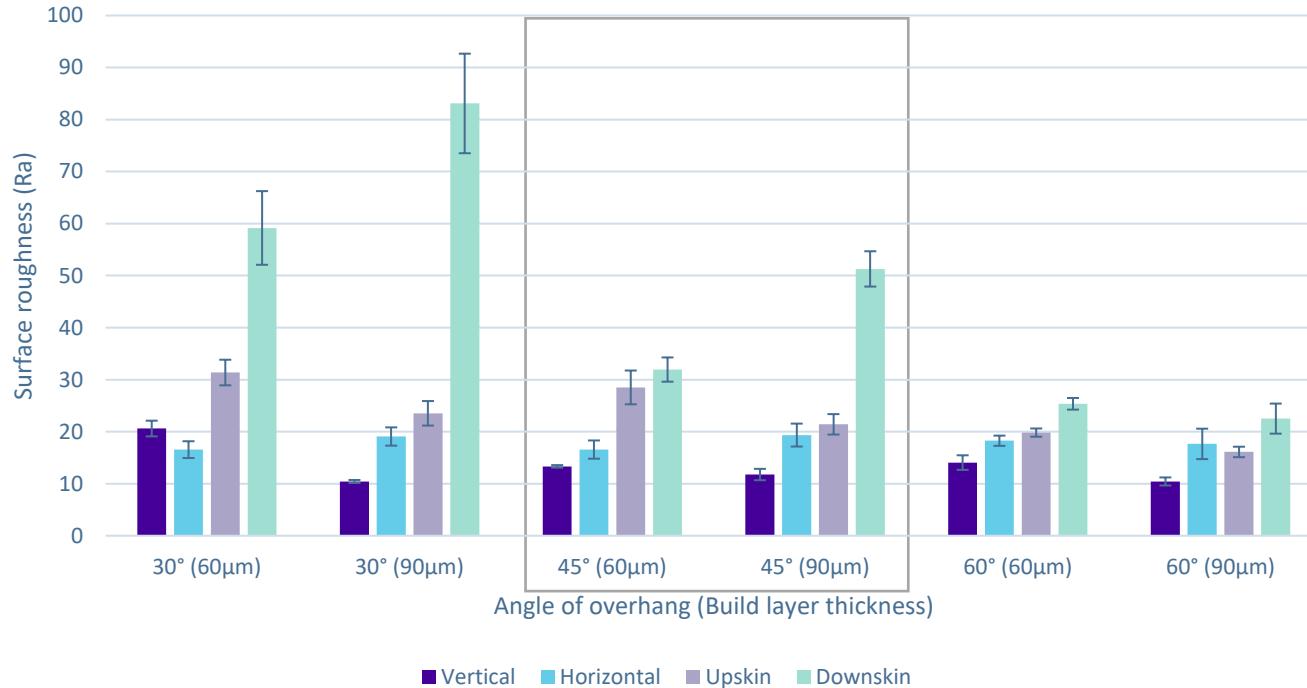


- There is little difference seen in the vertical, horizontal and upskin surface roughness measurements.
- The 60° blocks have the lowest surface roughness, particularly on the downskin.

4d. Surface roughness – 60 μm & 90 μm layer thickness



A comparison of the surface roughness measured in the 60 μm and the 90 μm layer thickness specimen blocks



- Surface roughness is increased by the 'stair-stepping' affect.
- Although in the overhang angle of 60°, the 90 μm layer thickness surface roughness is slightly lower than the 60 μm specimen blocks.

4d. Surface roughness – observations and conclusions

- In both the 60 μm and the 90 μm layer thickness surface roughness blocks, there is little difference seen in the vertical, horizontal and upskin surface roughness measurements.
- In both the 60 μm and the 90 μm layer thickness surface roughness blocks, the 60° blocks have the lowest surface roughness, particularly on the downskin.
- Surface roughness is increased by the 'stair-stepping' affect, although in the overhang angle of 60°, the 90 μm layer thickness surface roughness is slightly lower than the 60 μm specimen blocks.

Further work:

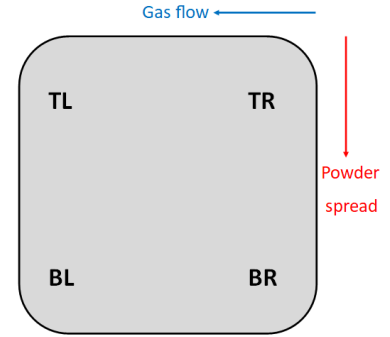
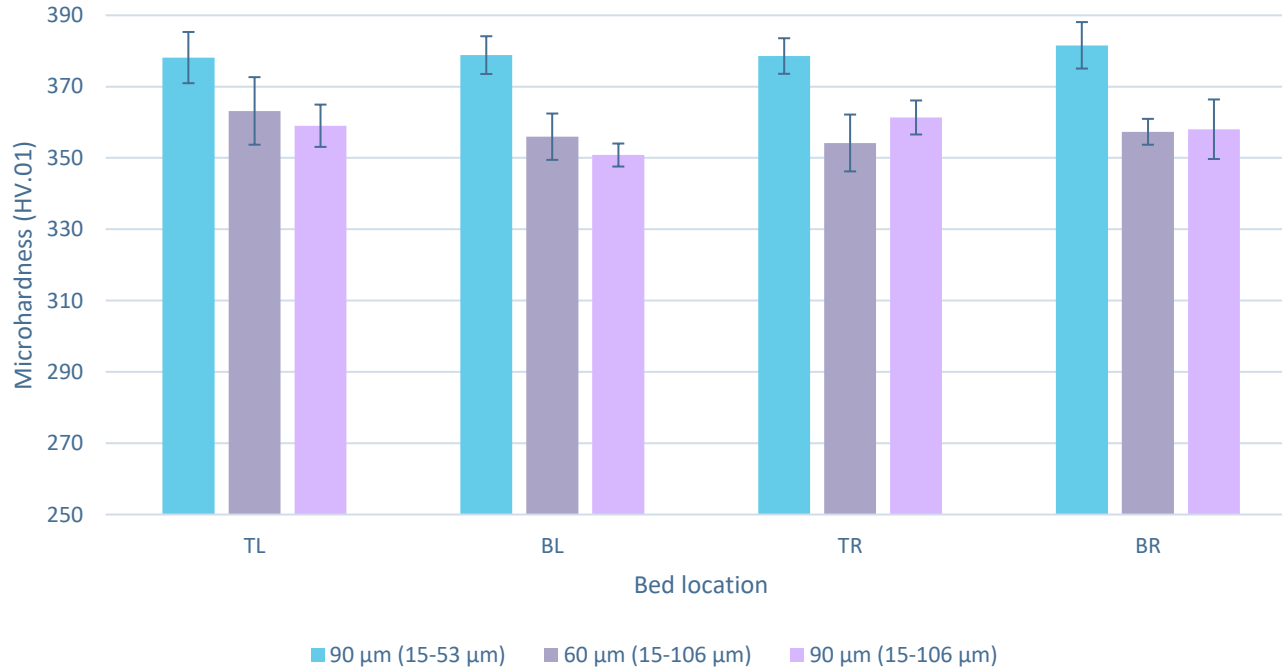
Optimisation of the contours/border scans may improve the surface quality.

4e. Microhardness - method

- To examine the effect of building in a coarser powder (15-106 μm) on the microhardness properties, indent measurements were carried out on cube specimens taken from the four quadrants of the build platform for each of three build conditions:
 - i. Standard powder (15-45 μm) in Thicker layers (90 μm)
 - ii. Coarser powder (15-106 μm) in Standard layers (60 μm)
 - iii. Coarser powder (15-106 μm) in Thicker layers (90 μm)
- Data for the Standard powder (15-45 μm) in Standard layers (60 μm) was taken from the Renishaw data sheet.
- One cube sample from each build location were sectioned in the as built (Z direction) using the Secotom, a desktop precision sectioning machine.. The samples were then hot mounted in edge-retentive Bakelite. The samples were ground with SiC paper (#220 grit), then polished with diamond grit (9 μm) and finished with 0.04 μm colloidal silica.
- Hardness assessment was conducted according to ASTM E384-17 in the bulk of each sample at 100g load using a Vickers indenter on a Struers Duramin-40 AC10. In total, six (6) individual measurements were conducted in the bulk of each sample and an average was calculated.

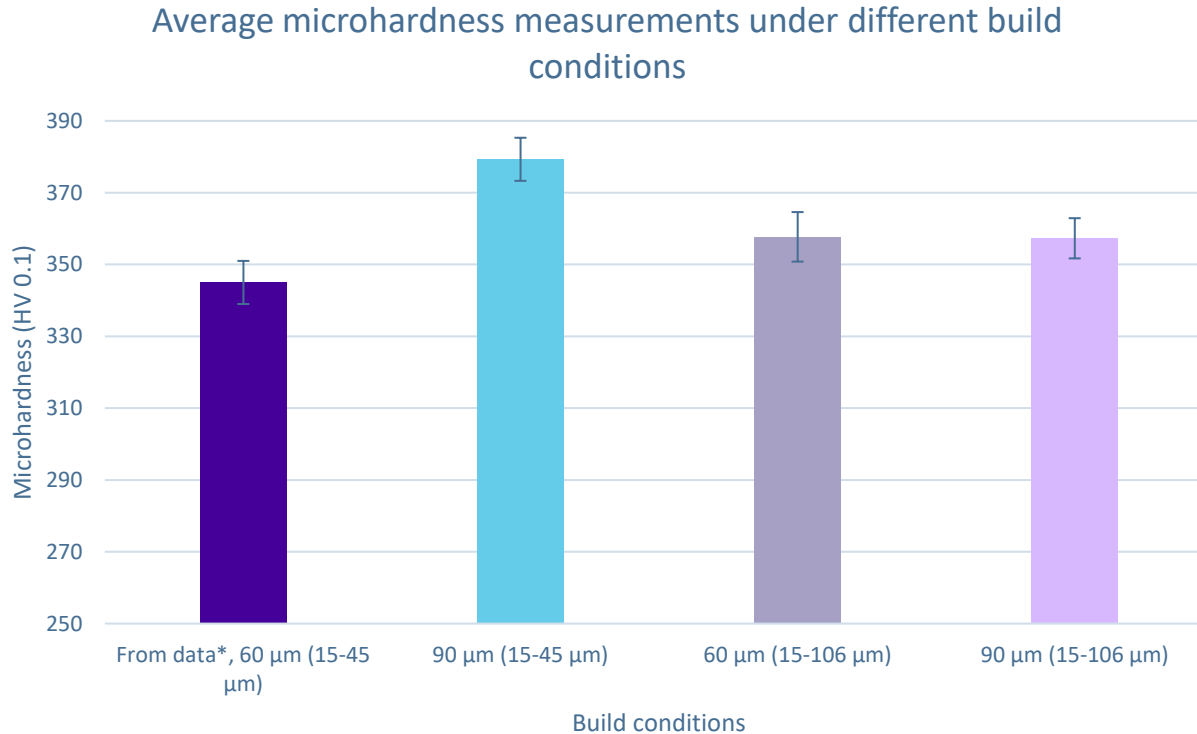
4e. Microhardness: a comparison

Microhardness measurements from different locations on the bed in different build conditions



There are six measurements from each build location on each specimen, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.

4e. Microhardness: a comparison



*Renishaw data is HV 0.5

- There are six measurements from each build location on each specimen and in this graph, the four build locations were averaged to compare the different build conditions.
- The standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.

4e. Microhardness – observations and conclusions

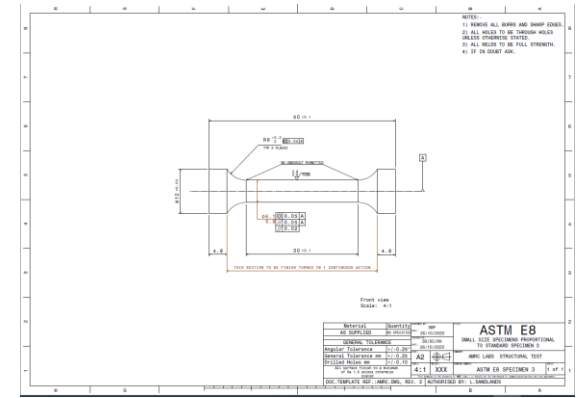
- There appears to be no significant difference in the microhardness measurements between the different build locations in the specimens produced.
- The biggest differences were seen in the coarser powder (15-106 μm PSD) when comparing the 60 μm results with those built in thicker layers (90 μm), however, overlapping error bars means that the difference is not significant.
- In the coarser powder, microhardness results were very consistent despite the different layer thickness: 357.7 HV 0.1 (60 μm layers) and 357.3 HV 0.1 (90 μm layers) on average.
- In the standard powder, microhardness results were significantly higher in the thicker layers at 379.3 HV 0.1 (90 μm layers) and 345 HV 0.5 in the standard layers (60 μm layers) (taken from the Renishaw data sheet).

Further work:

- Enhanced tensile strength and microhardness are the result of a finer grain structure, but will result in poorer ductility as hardness of Ti-6AL-4V is dependent on the grain size, specifically the thickness of the α laths. Further analysis of the microstructure could show the reason for the higher hardness measurements in the Standard powder (15-53 μm) in the Coarser layers (90 μm).

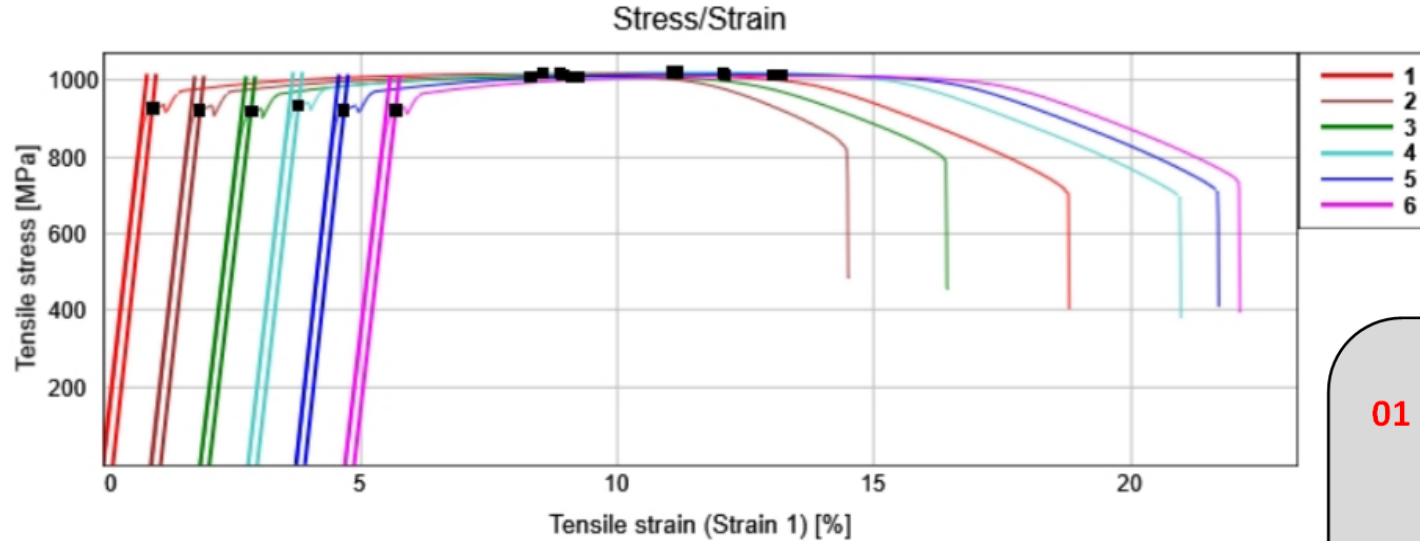
4f. Tensile properties – method

- To examine the effect of building in a coarser powder (15-106 μm) on the tensile properties, specimens were manufactured in the two different layer thicknesses in the coarser powder:
 - Coarser powder (15-106 μm) in Standard layers (60 μm)
 - Coarser powder (15-106 μm) in Thicker layers (90 μm)
- Data for the **Standard powder (15-45 μm) in Standard layers (60 μm)** was taken from the Renishaw data sheet.
- The tensile tests were conducted according to ASTM E8M, each specimen was inspected before the test. Tensile testing was carried out to the in-house laboratory test procedure ASTC.TP12 Static Tensile Testing of Metallic Materials, Rev. 17 (Oct. 2021). The gauge length was marked on the test specimen before inserting it into the Instron 50kN test frame. Control Method C was used to determine the yield and UTS properties up to the point of specimen fracture. The test specimen was removed from the test frame, placed back together carefully, and the elongation and reduction in area measurements were taken and entered into the test programme.

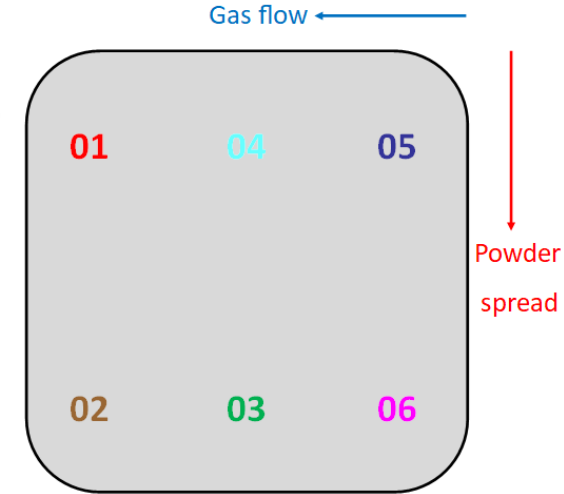


4f. Tensile properties

i. Coarser powder (15-106 μm) in Standard layers (60 μm)

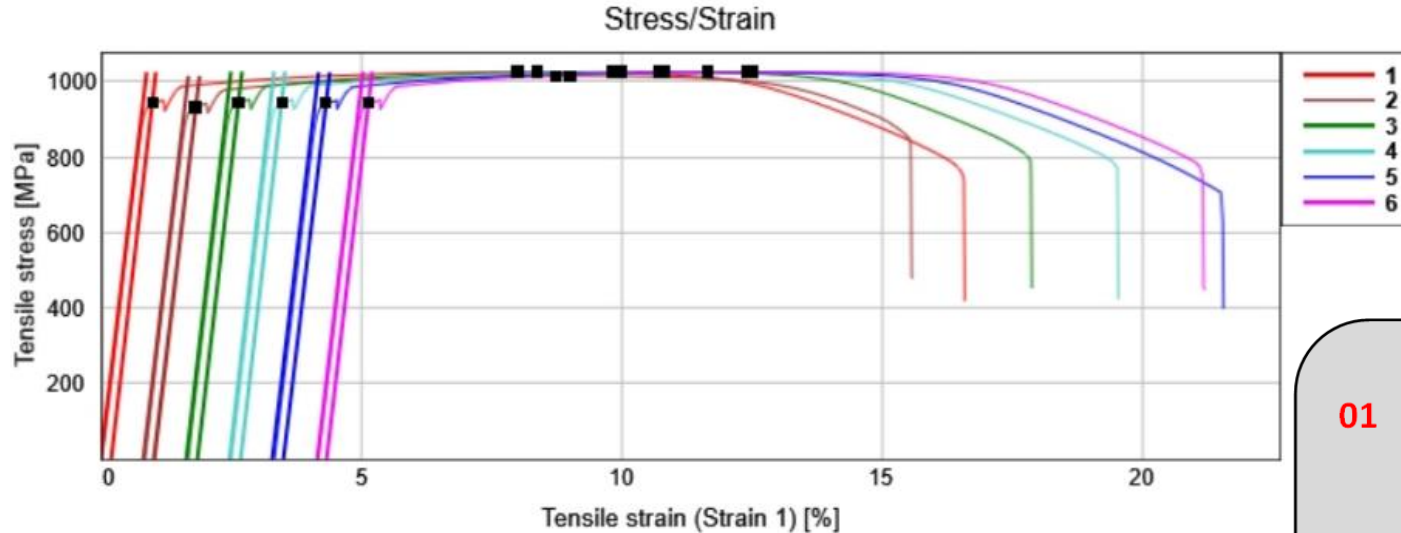


- Poorest performance observed in the bottom-left corner – furthest from the gas outlet and powder spread and area that suffers the most from spatter being redeposited.

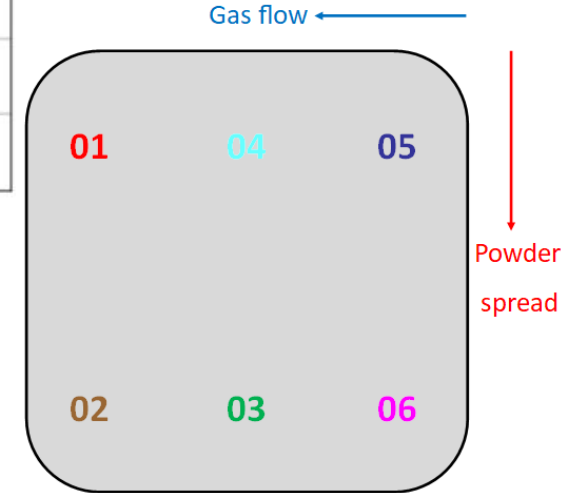


4f. Tensile properties

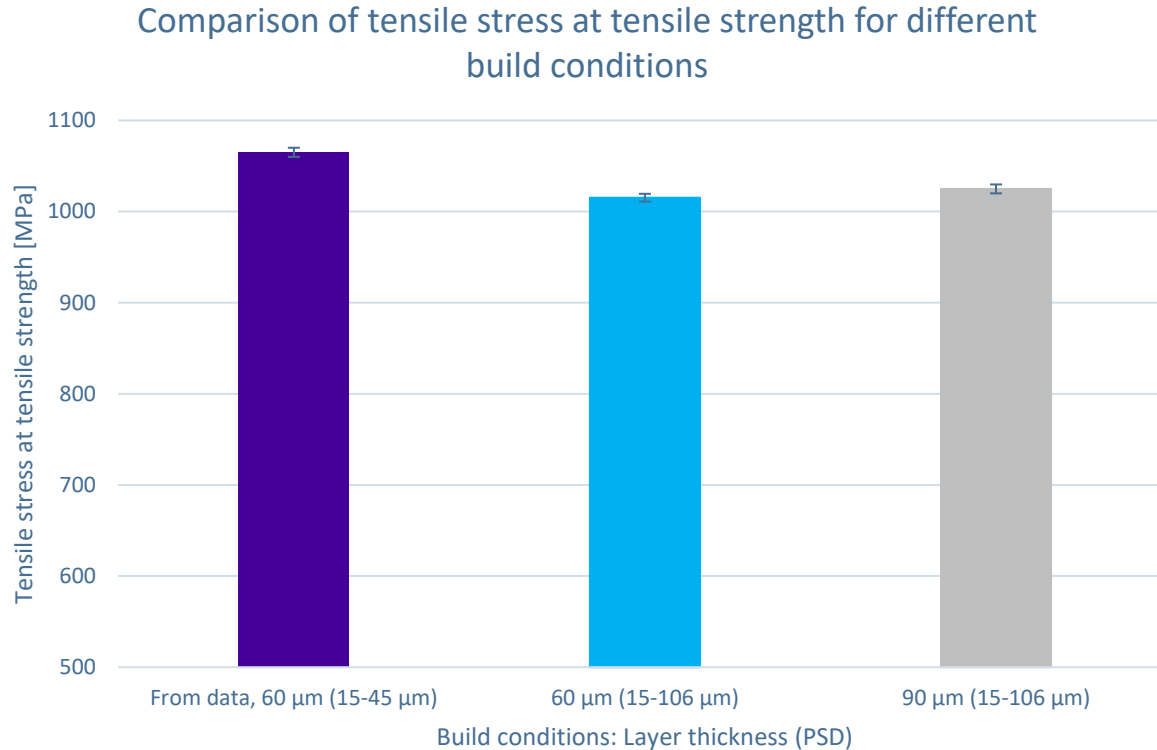
ii. Coarser powder (15-106 μm) in Thicker layers (90 μm)



- Poorest performance observed in the bottom-left corner – furthest from the gas outlet and powder spread and area that suffers the most from spatter being redeposited.
- Overall performance better than the Standard (60 μm layers).



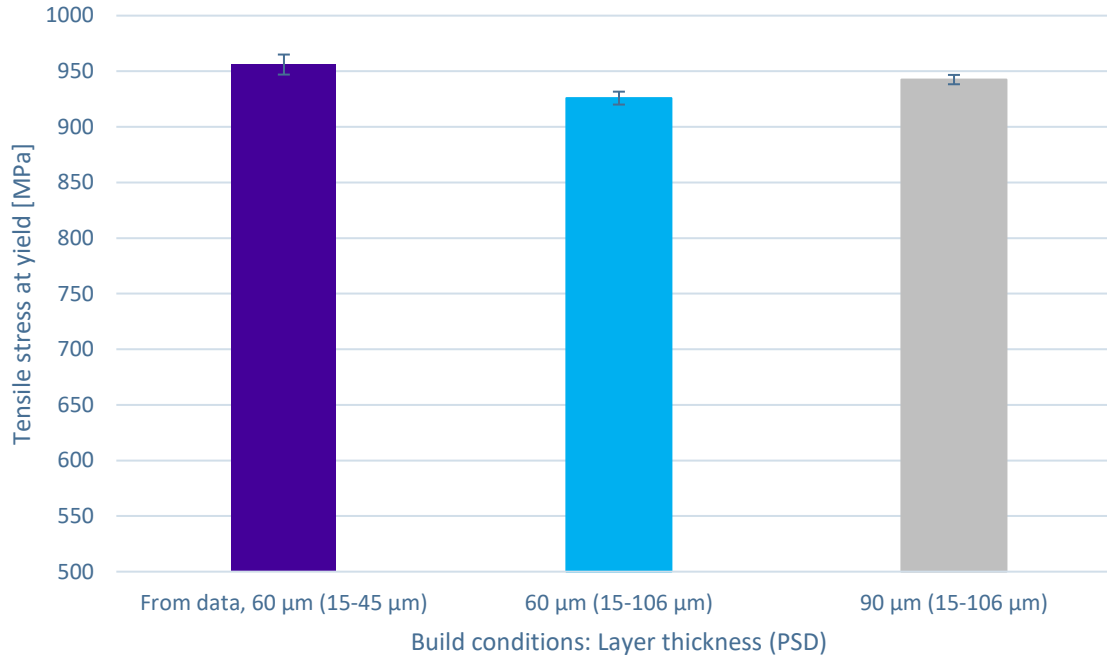
4f. Tensile properties



- The six tensile results from across the build were averaged and plotted.
- It shows a small reduction in tensile strength when compared to the performance of specimens produced in the standard powder in standard layers (as taken from the data sheet).
- The thicker layer specimens (90 μm) had a slightly higher overall tensile strength (1028 MPa) than the standard layer (60 μm) specimens (1015 MPa).

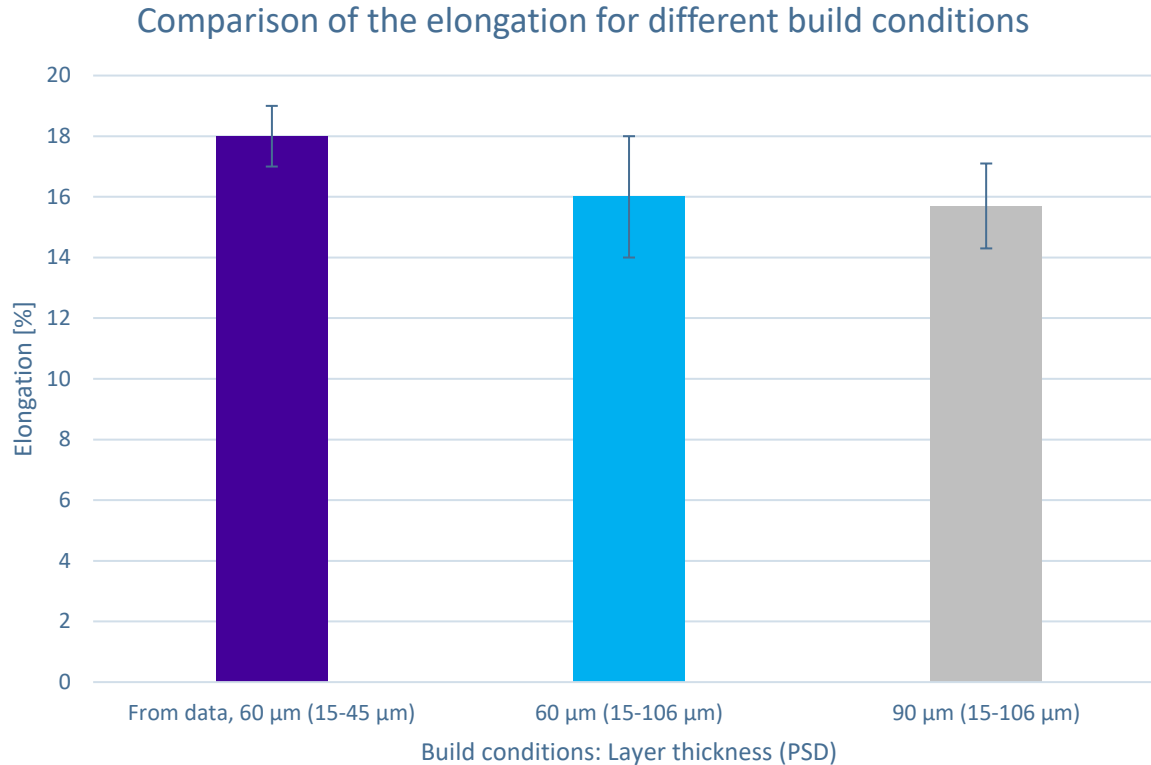
4f. Tensile properties

Comparison of yield stress at yield for different build conditions



- The six tensile results from across the build were averaged and plotted.
- The thicker layer (90 μm) specimens (943 MPa) performed close to the standard conditions (956 MPa).
- The thicker layer specimens (90 μm) had a slightly higher overall yield strength (943 MPa) than the standard layer (60 μm) specimens (925 MPa).

4f. Tensile properties



- The six elongation measurement results from across the build were averaged and plotted.
- Due to the overlapping of the error bars, there is no significant difference between the different conditions.
- The specimens built in standard layers (60 μm) the coarser powder had an elongation of 16 % and the thicker layer (90 μm) specimens were 15.7 % on average.

4f. Tensile properties – observations and conclusions

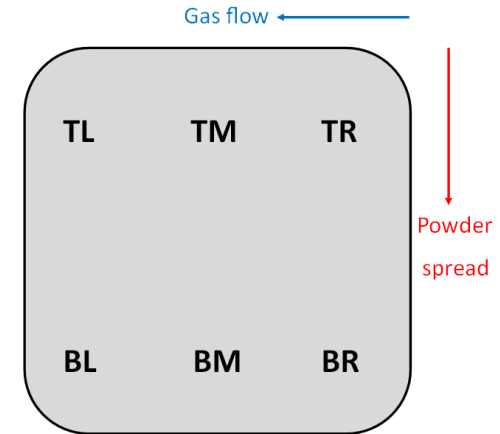
- Build location has an impact on the mechanical performance of the specimens produced, with the left side of the build and specifically the left-bottom area performing the worst.
- In terms of the tensile strength, the yield stress and the elongation, the results from the data sheet for the standard powder and the standard layers are slightly better than the specimens produced in the coarser powder, however the difference is minimal and not significant.
- For both the tensile strength and the yield stress, the results were very slightly better for the thicker layer (90 μm) specimens than the standard layer (60 μm) specimens, but the difference is not significant.
- For the elongation results, the standard layer (60 μm) specimens performed slightly better (16 %) than the thicker layer (90 μm) specimens (15.7 %), although not significantly.

Future work:

- Build the sample set in the standard powder on this machine and test the same bars for a true comparison.
- Fracture analysis of the cause of the breaks.

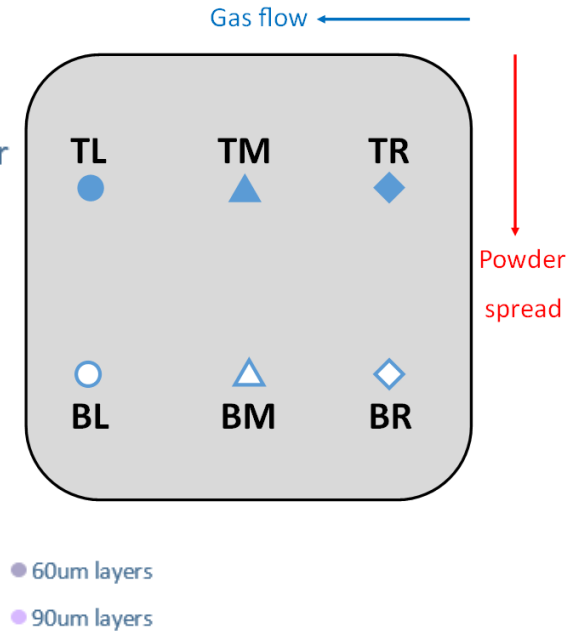
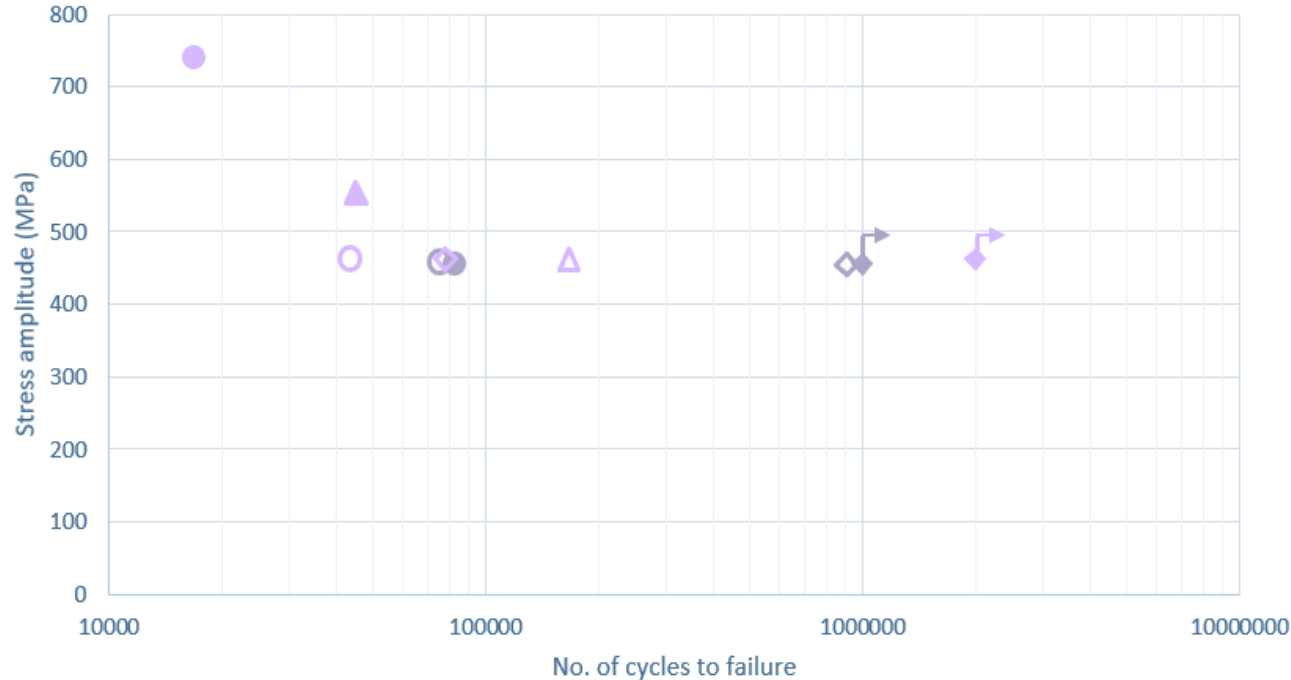
4g. Fatigue - method

- Fatigue testing was carried out on an Instron 100kN servo hydraulic universal testing machine.
- The fatigue tests were conducted following an in-house laboratory test procedure (ASTC TP09-02). This details the handling, machine setup, instrumentation and calibration, leading to the subsequent testing of ASTM E606-E606M fatigue specimens in force control. Load levels and completed test cycles were recorded.
- For the builds carried out in the coarser powder (15-106 μm) and built in 60 μm and 90 μm layers, specimens were taken from the four quarters of the build plate (top left, bottom left, top right and bottom right), to determine if there are any differences caused by build location.
- For the 90 μm layer thickness build, an additional two samples were tested, using specimens taken from the top middle and bottom middle locations on the built plate.



4g. Fatigue – results

Comparison of low-cycle fatigue life of specimens built in the coarser powder (15-106 μm) in 60 μm and 90 μm layers.



The  arrows indicate that the specimen continued and did not fail.

4g. Fatigue – observations and conclusions

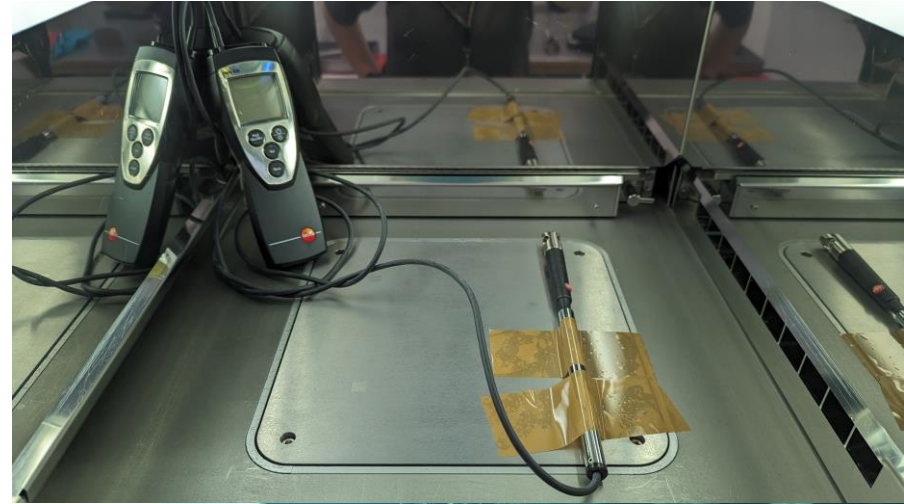
- At 50 % of the yield stress, the bottom left specimens for both 60 and 90 μm layer thicknesses test bars failed before 100,000 cycles.
- At 50 % of the yield stress, the top right specimens for both 60 and 90 μm layer thicknesses test bars did not fail and continued past 1,000,000 cycles.
- Specimens from the bottom left consistently showed lower fatigue life compared to other locations, suggesting that factors such as build orientation and possible stress concentrations or defects at this location affect the performance. This draws the conclusion that while there are differences in the fatigue life between 60 μm and 90 μm layer thicknesses, the build location appears to have a more pronounced effect.

Future work:

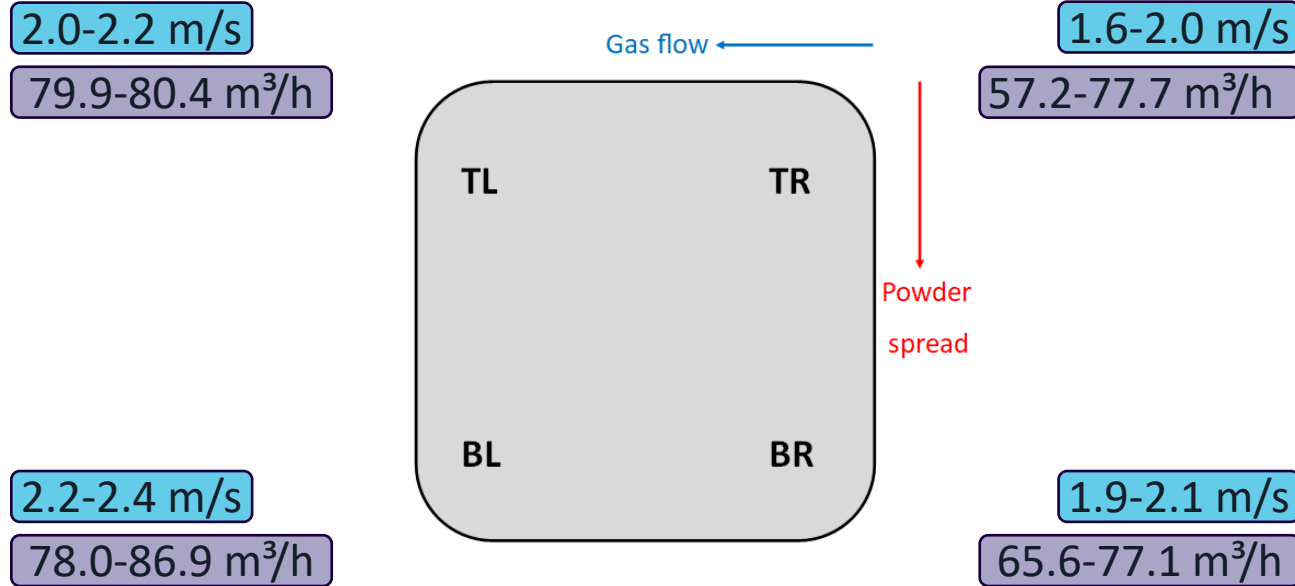
- Build the sample set in the standard powder on this machine and test the same bars for a comparison of the feedstock powders.
- Fracture analysis of the cause of the breaks, looking specifically for any initiation sites, porosity or inclusions that could indicate the cause of the failures.

4h. Gas flow conditions – method

- It appears from the tensile results that the build location has an impact on the mechanical performance of the specimens produced, with the left side of the build and specifically the left-bottom area performing the worst.
- Using an anemometer, measurements were taken in the four (4) corners of the build platform to see if there is a difference in the gas flow between those regions.
- The pump is set at $190 \text{ m}^3/\text{h}$ – this the argon flow rate that is recommended for Ti-6Al-4V for this machine by Renishaw.
- The top number on the readout is the speed (m/s) and the bottom number is the volume rate (m^3/h).



4h. Gas flow conditions – results



- Once the argon flow rate was at 190 m³/h (as read from the PLC on the Ren500Q), the minimum and the maximum speed and volume rate was recorded for each bed location.
- The readings show that the gas flow rate is higher on the left side, away from the gas outlet and highest overall in the bottom-left location.

4h. Gas flow conditions – observations and conclusions

- It is important in PBF-LB for the gas flow to be optimised in such a way that it can carry away spatter that is ejected from the melt area whilst ensuring that it does not blow away the powder that is being spread by the recoater for each layer. A good balance is important.
- It is unclear as to whether the gas flow for the coarser powder (15-106 μ m) range needs to be altered to compensate for the increased spatter ejected, or whether larger spatter particles are produced and whether the standard gas flow rate is sufficient to effectively remove the spatter. It is worth noting that the maximum argon flow rate that this AM machine can be set to is 200 m³/h, therefore there is little room for an increase.
- The readings from the anemometer show that the bottom-left area of the build platform experiences the highest gas flow rate when compared to the other bed locations and it is in this area that tensile and fatigue performance is at it's lowest.

Further work:

- Further investigations on the significance of a varied gas flow on the resulting part performance dependent on the bed location.

5. Laser parameter development: Can the spatter output be reduced?

5a. Design of experiments: Lowering the scan speed and the laser power.

5b. Density results.

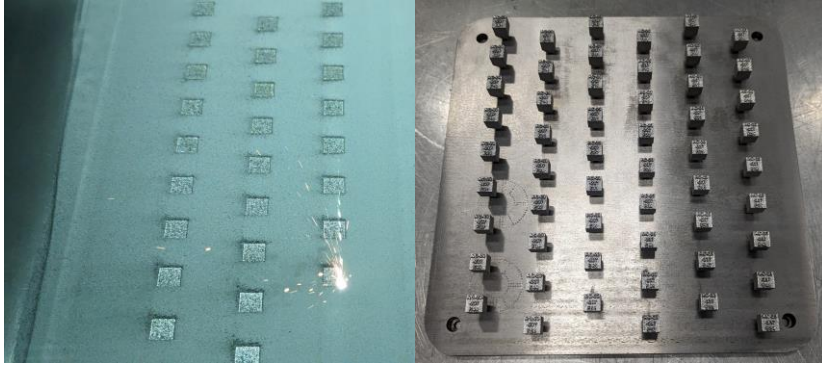
5c. Microhardness results.

5d. Tensile results.

5e. Surface roughness results.

5. Laser parameter development: Can the spatter output be reduced?

- A design of experiments, taking the set of parameters used previously, but lowering the power and the scan speed to attempt to stabilise the melt pool, reduce spatter and improve mechanical performance.



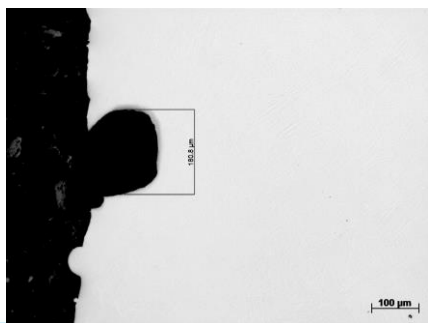
- It could reduce the productivity when compared with the previous set of parameters, but may improve the mechanical properties and the length of the life of the power by reducing spatter.
 - Three (3) repeats of each set of parameters were printed in order to examine the effects of:
 - i. Lowering the scan speed
 - ii. Lowering the power
 - iii. Lowering the scan speed and the power
- The specimens, manufactured using the different parameters, were examined with results obtained for:
 - 5b. Density (cubes sectioned in the as-built direction).
- Following the isolation of the parameters that printed specimens with the best density, a full test build was carried out to examine the effects on:
 - 5c. Microhardness
 - 5d. Tensile
 - 5e. Surface roughness

5a. Design of experiments - method

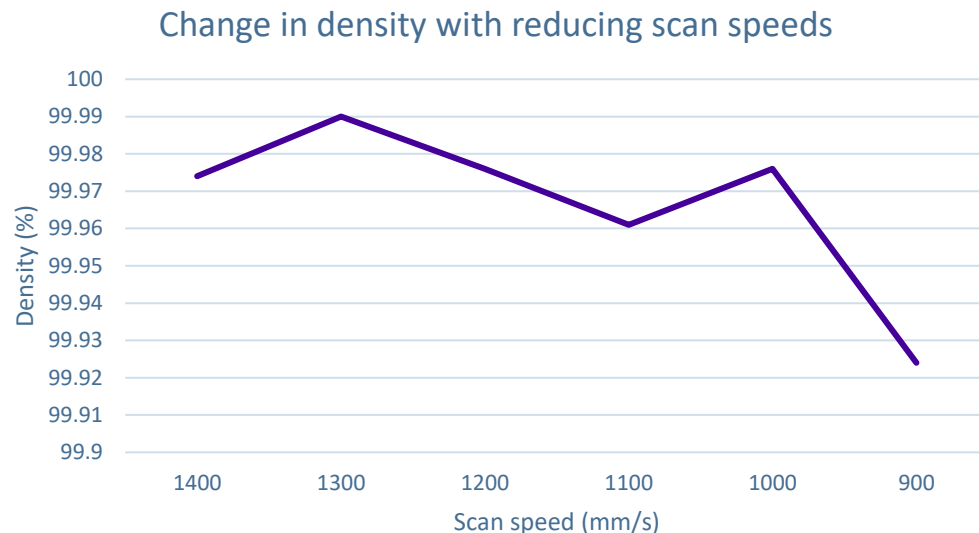
i. Lowering scan speed

- Set of cubes reducing the scan speed only, power remained at 425 W and hatch distance at 0.14 mm.
- It was found that density dropped with reducing scan speeds.

Sample number	Scan speed (mm/s)	VED (J/mm ³)	Density (%)
1	1400	24.09	99.974
2	1300	25.95	99.990
3	1200	28.11	99.976
4	1100	30.66	99.961
5	1000	33.73	99.976
6	900	28.11	99.924



Sample 6: Surface finish (edge of cubes) had large gaps/pores across the whole length.

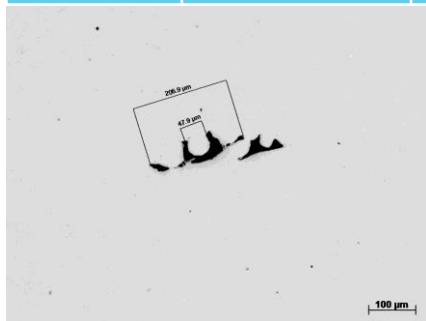


5a. Design of experiments - method

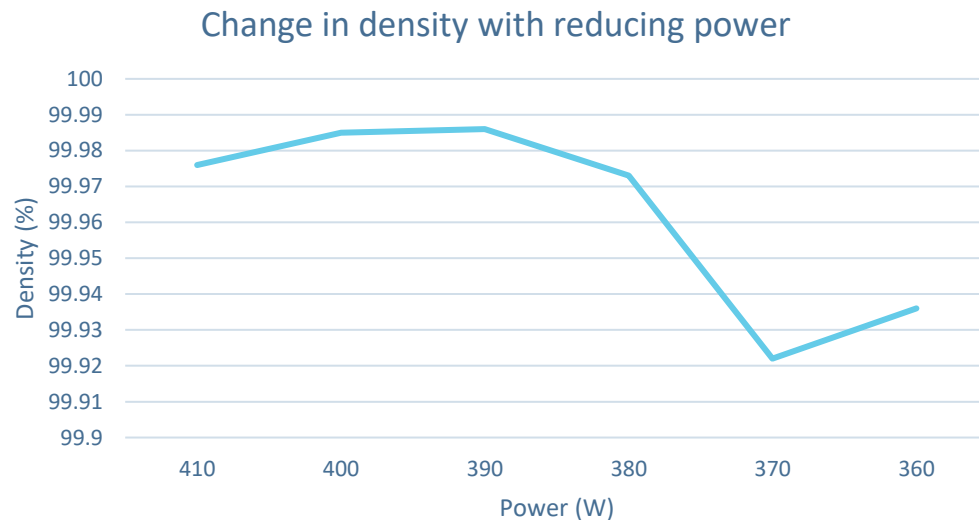
ii. Lowering the power

- Set of cubes reducing the power only, scan speed remained at 1400 mm/s & hatch distance at 0.14 mm.
- It was found that density dropped with reducing the power, with increasing lack-of-fusion defects.

Sample number	Power (W)	VED (J/mm ³)	Density (%)
7	410	23.24	99.976
8	400	22.67	99.985
9	390	22.11	99.986
10	380	21.54	99.973
11	370	20.98	99.922
12	360	20.41	99.936



Sample 11: Lack-of-fusion defects seen in samples with lower power.

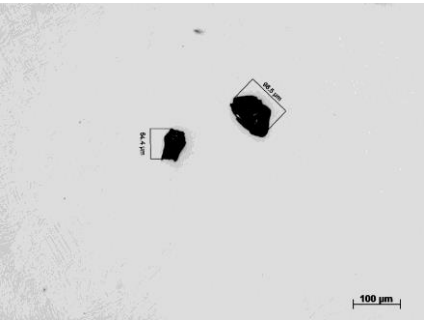


5a. Design of experiments - method

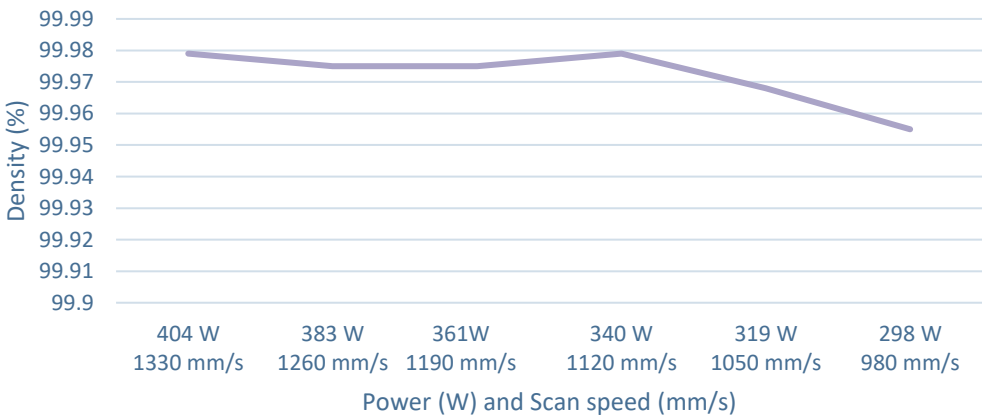
iii. Lowering the power & scan speed

- Set of cubes reducing the power and the scan speed, keeping the hatch distance at 0.14 mm.
- It was found that density stayed steady until the power dropped to around 320 W with a scan speed of 1050 mm/s.

Sample number	Power (W)	Scan speed (mm/s)	VED (J/mm ³)	Density (%)
13	404	1330	24.09	99.979
14	383	1260	24.09	99.975
15	361	1190	24.09	99.975
16	340	1120	24.09	99.979
17	319	1050	24.09	99.968
18	298	980	24.09	99.955



Change in density with reducing power and scan speed



Sample 18: Larger pores observed with the lowest power and scan speed.

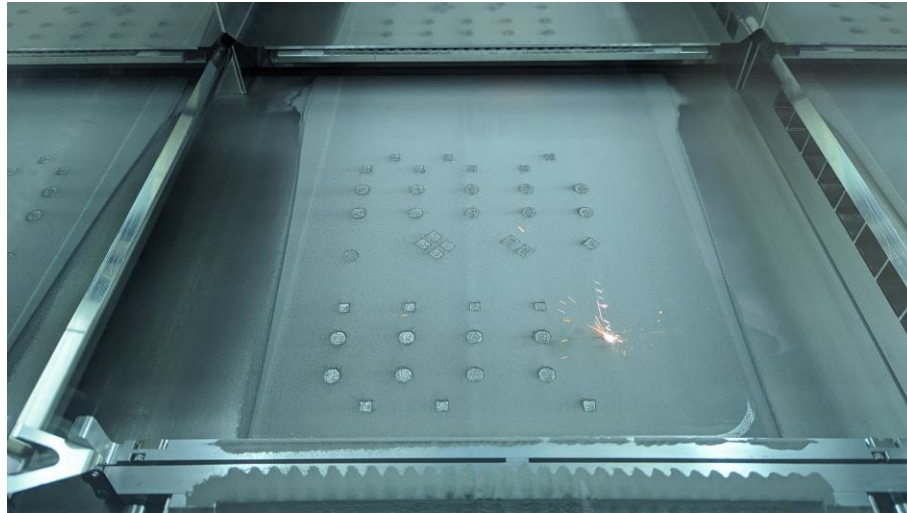
5a. Design of experiments - method

- Following density analysis of all cubes, the set used to build sample 16 were used to manufacture a full test build.

Sample number	Power (W)	Scan speed (mm/s)	VED (J/mm ³)	Density (%)
1	425	1400	24.09	99.974



Sample number	Power (W)	Scan speed (mm/s)	VED (J/mm ³)	Density (%)
16	340	1120	24.09	99.979



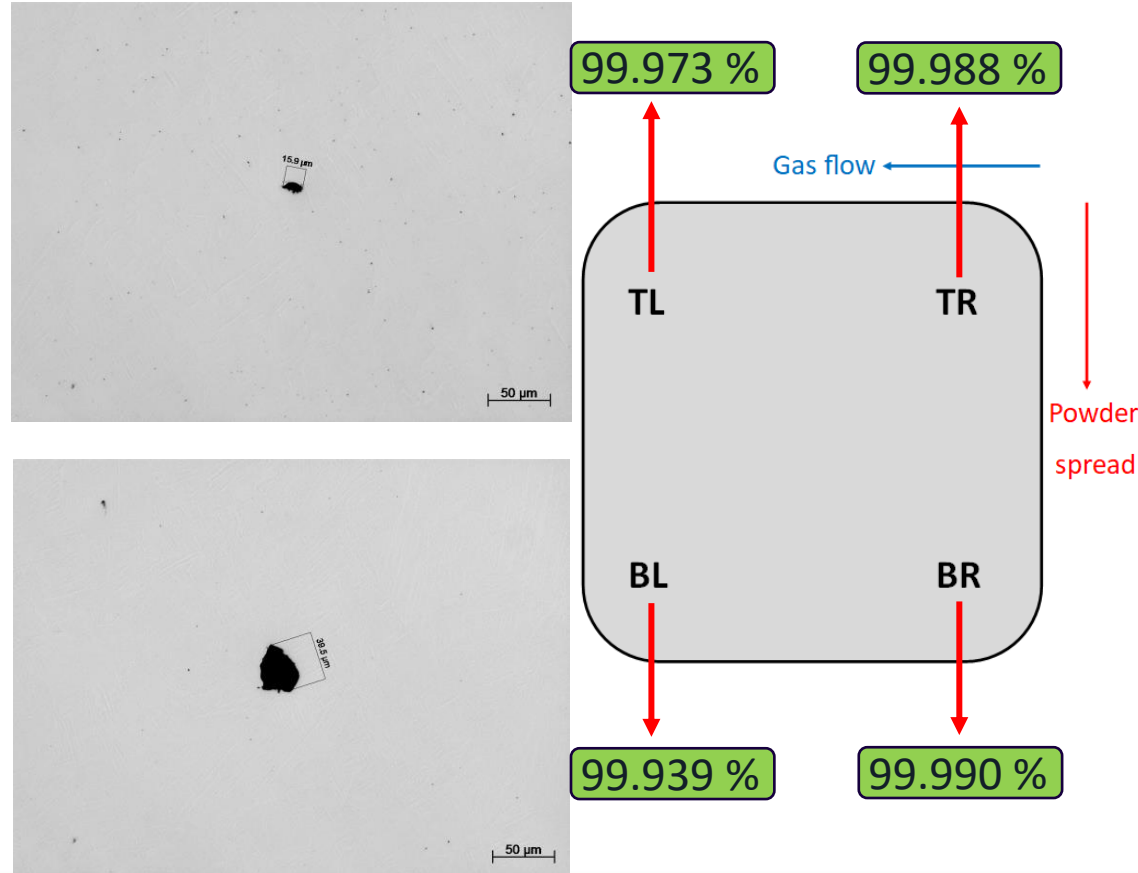
- These parameters slow the build down from 14 hours 15 minutes to 16 hours 18 minutes.
- Spatter is still observed on the left side of the specimens as they are building.

5b. Density results

- To examine the effect of reducing the power and the scan speed on the density, measurements were carried out on cube specimens taken from the four quadrants of the build platform.
- One cube sample from each build location were sectioned in the as built (Z direction) using the Secotom, a desktop precision sectioning machine. The samples were then hot mounted in edge-retentive Bakelite. The samples were ground with SiC paper (#220 grit), then polished with diamond grit (9 μ m) and finished with 0.04 μ m colloidal silica. For the assessment of the density, the samples were imaged using the Zeiss Axio microscope with the porosity (density) being measured using the built-in software.
- The specimens were built with **Coarser powder (15-106 μ m)** in **Thicker layers (90 μ m)** but with the parameters that dropped the power from **425 to 340 W** & the scan speed from **1400 to 1120 mm/s**.
- The following slide shows higher magnification (x200) images of two of the build locations as an example and the measured density for each of the four (4) build locations.

5b. Density results

Power = 340 W & Scan speed = 1120 mm/s

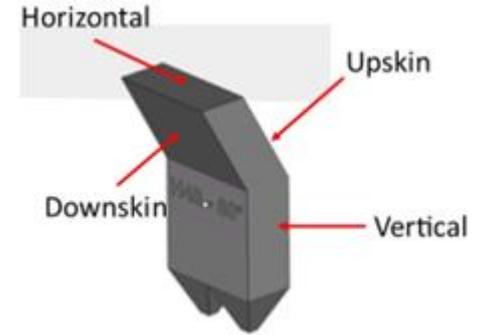


- Shown here are two examples of the largest pores observed across the four specimens taken from the four build platform locations.
- The density with this new set of parameters is the same as with the higher power and higher scan speed: on average, the density for the higher power/scan speed = 99.973 %, with the lower power/scan speed = 99.972 %.

5c. Surface roughness results

Power = 340 W & Scan speed = 1120 mm/s

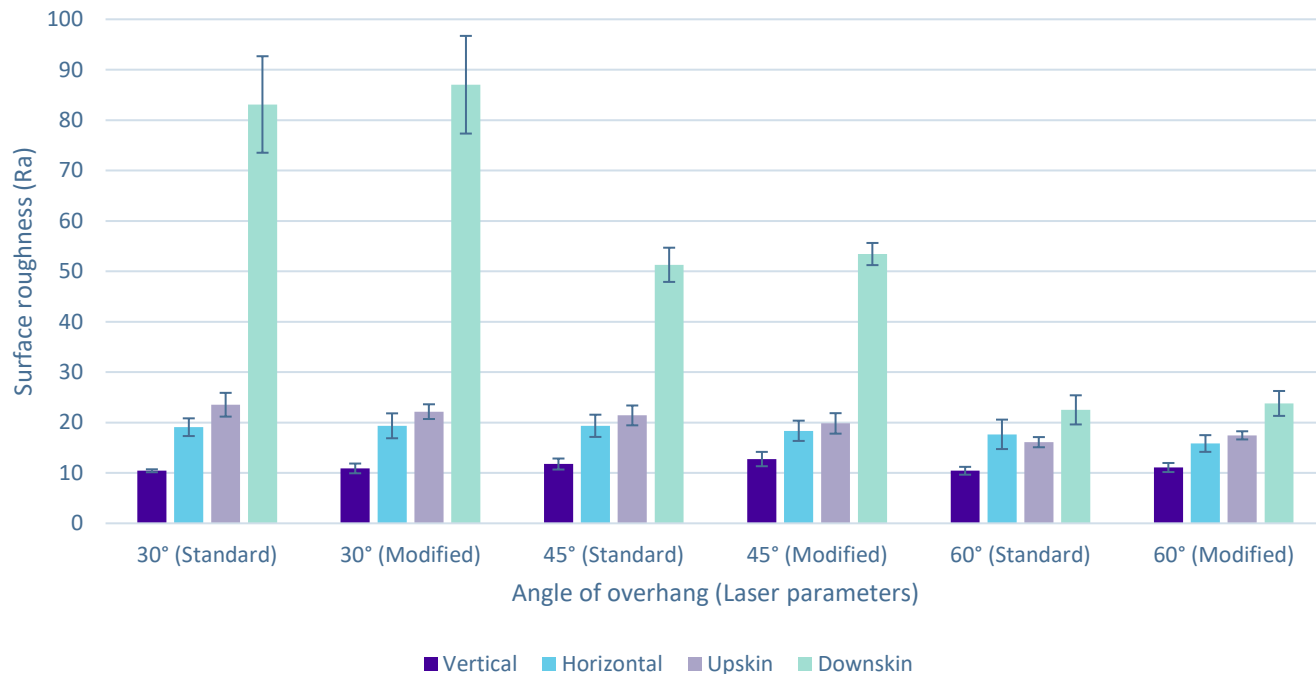
- In order to assess the affect of the coarser powder on the surface roughness with the modified laser parameters, blocks were printed with the three different overhang angles: 60°, 45° and 30° and the upskin, the downskin, the vertical and the horizontal surfaces were measured.
- The measurements were carried out on a Keyence microscope due to the high surface roughness of the specimens.
- Five random locations on each of the four surfaces were imaged and measured and averaged, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.



5c. Surface roughness results

Power = 340 W & Scan speed = 1120 mm/s

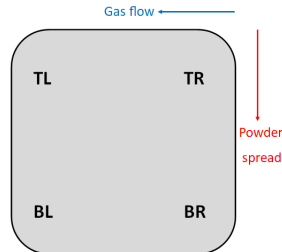
A comparison of the surface roughness between the standard and the modified laser parameters measured in the 90 μm layer thickness specimen blocks



- The modified laser parameters show no improvement to the surface roughness when compared to the specimens printed in the standard laser parameters.

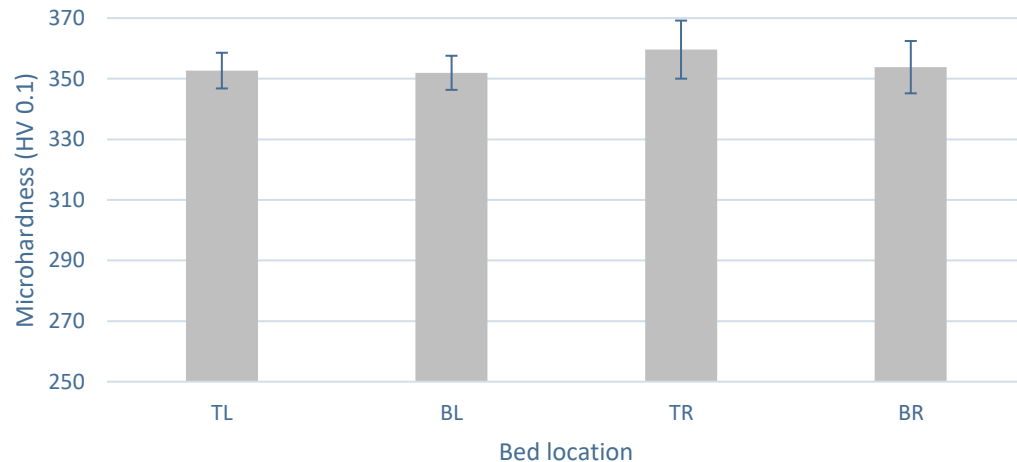
5d. Microhardness results

Power = 340 W & Scan speed = 1120 mm/s

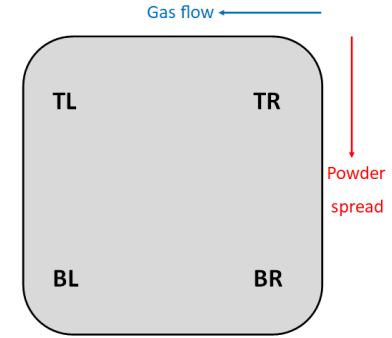
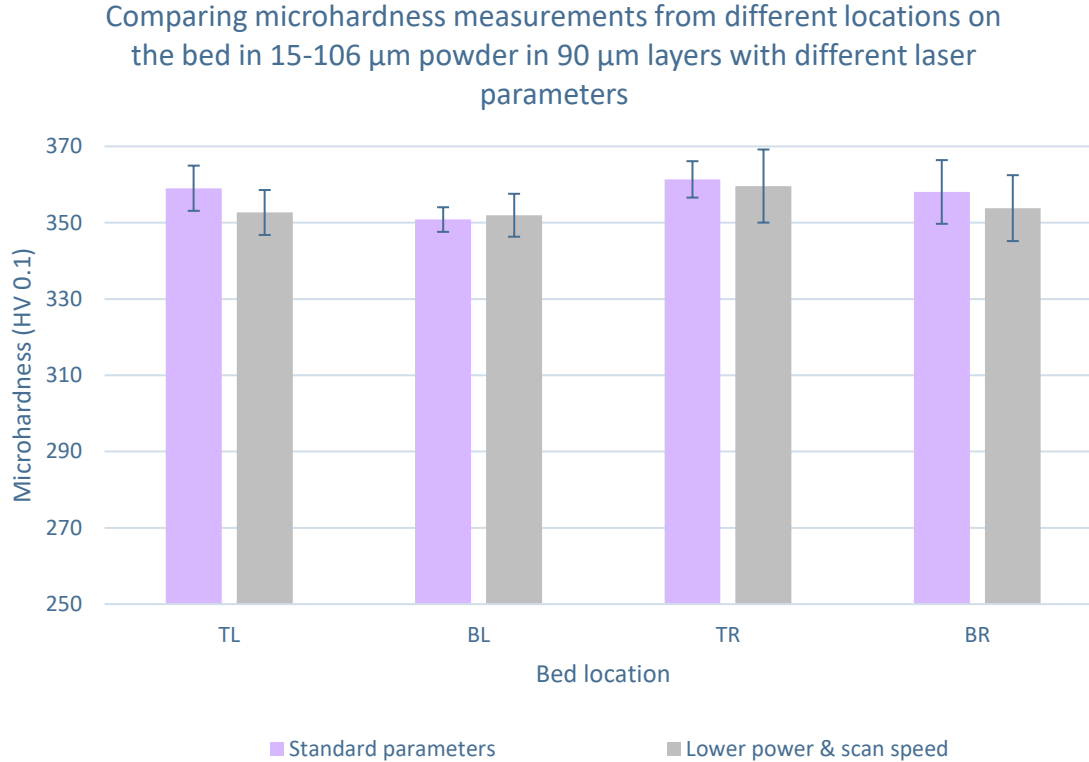


- To examine the effect of reducing the power and the scan speed on the microhardness, measurements were carried out on cube specimens taken from the four quadrants of the build platform on the specimens prepared as previously described.
- Hardness assessment was conducted according to ASTM E384-17 in the bulk of each sample at 100g load using a Vickers indenter on a Struers Duramin-40 AC10. In total, six (6) individual measurements were conducted in the bulk of each sample and an average was calculated.

Microhardness measurements from different locations on the bed in 15-45 μm powder in 90 μm layers with lower power (340 W) and scan speed (1120 mm/s)



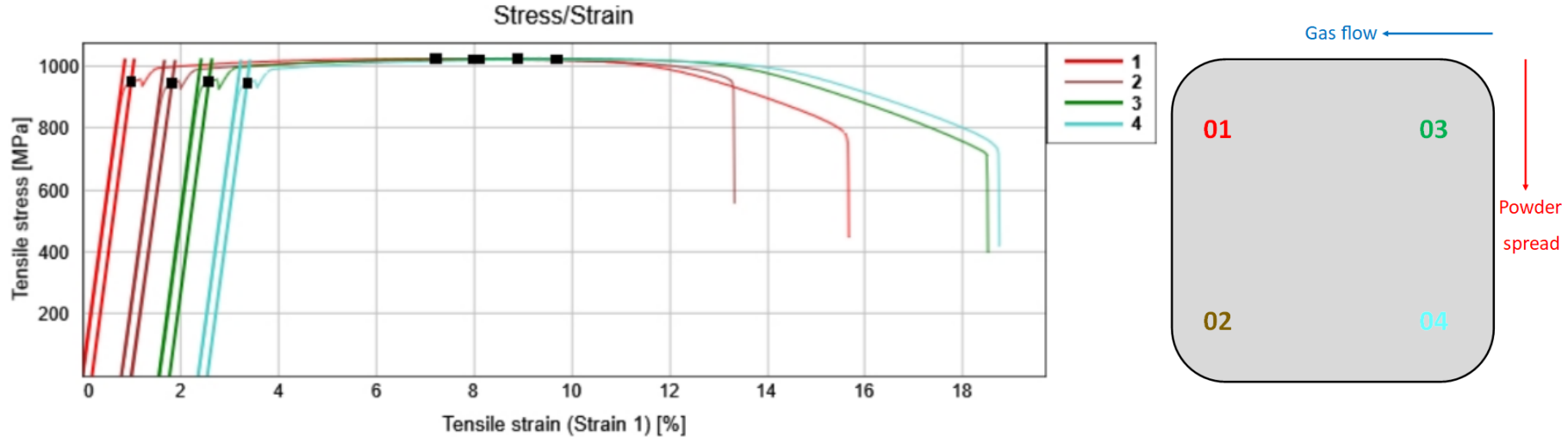
5d. Microhardness results



- When comparing the microhardness between the higher power/scan speed and the lower power/ scan speed, there is no significant difference between the results.

5e. Tensile results

Power = 340 W & Scan speed = 1120 mm/s

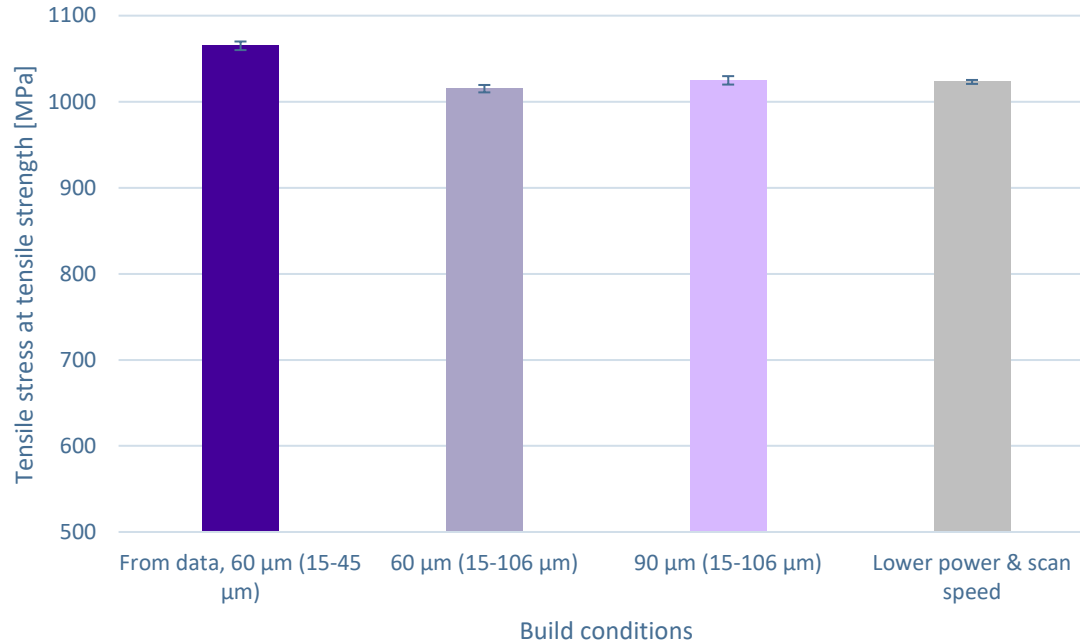


- To examine the impact to the tensile properties of building in a coarser powder (15-106 μm) with the modified power and scan speed parameters, specimens were manufactured in a 90 μm layer thickness.
- The tensile tests were conducted according to ASTM E8M, the method has been described previously.
- Poorest performance observed in the bottom-left corner – furthest from the gas outlet and powder spread and area that suffers the most from spatter being redeposited.

5e. Tensile results – Tensile strength

Power = 340 W & Scan speed = 1120 mm/s

Comparison of tensile stress at tensile strength for different build conditions

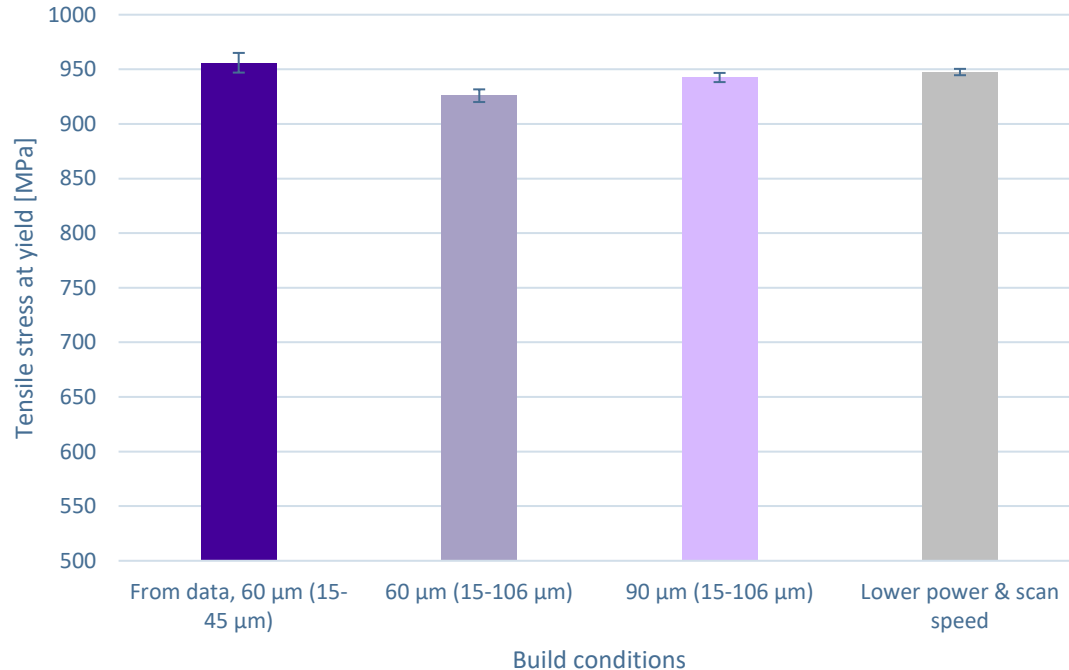


- The four tensile results for strength, yield stress and elongation (see next two slides) from across the build were averaged and plotted to compare with the other build conditions.
- Due to the overlapping of the error bars, there is no significant difference between the standard parameters and the parameters with a lower power and lower scan speed.
- However, the specimens built in the modified parameters show no improvement in the tensile strength over those built in the higher power and scan speed.

5e. Tensile results – Yield stress

Power = 340 W & Scan speed = 1120 mm/s

Comparison of tensile stress at yield for different build conditions

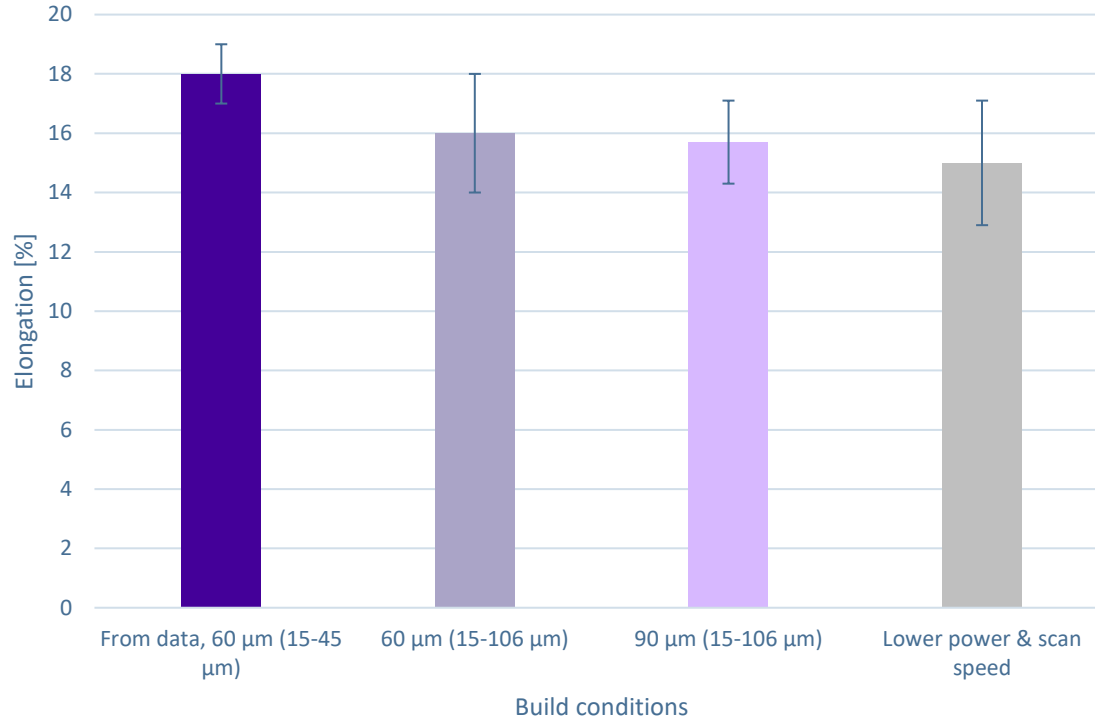


- The specimens built in the modified parameters show no improvement in the yield stress over those built in the higher power and scan speed.

5e. Tensile results – Elongation

Power = 340 W & Scan speed = 1120 mm/s

Comparison of the elongation for different build conditions



- The specimen from the BL area of the build plate had just 12 % elongation and pulled down the average, resulting in a worse performance on average compared to the standard parameters.

5. Laser parameter development – observations and conclusions

Power = 340 W & Scan speed = 1120 mm/s

- A DoE was carried out to ensure a new set of parameters would not adversely affect the density and a set with lower power and scan speed was found to achieve low porosity.
- A full build in the refined parameters did not appear to reduce the spatter output (visual observation only).
- When comparing the microhardness between the higher power/scan speed and the lower power/scan speed, there is no significant difference between the results.
- The specimens built in the modified parameters show no improvement in the tensile strength, the yield stress or elongation over those built in the higher power and scan speed.
- The lack of improvement in mechanical performance and the extended build time due to the reduced scan speed means this approach is not economically beneficial.

Further work:

- An examination of the microstructure using the SEM and how it affects the mechanical performance.
- A fracture analysis of the breaks in the tensile bars could determine what caused the early failure.

6. Demonstrator test case: Atherton Bikes

Building a full build plate of bike lugs in Coarser powder (15-106 μm) in both Standard layers (60 μm) and Thicker layers (90 μm)

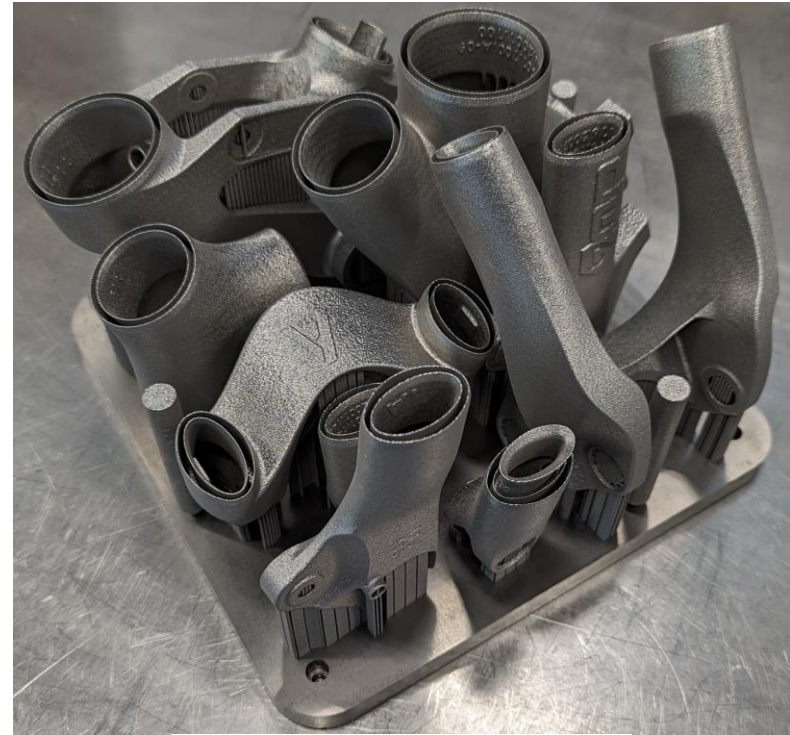
6. Demonstrator parts: Atherton Bikes

Problems for Atherton Bikes in using PBF-LB to manufacture parts:

- They have a high demand to produce their lug components and have therefore switched to running 24 hour shifts.
- There is little to no profit margin on AM components and are looking to reduce costs and increase production rates. The current build time is 20 hours in 60 μm layers to produce each build plate which contains all the parts required to build one bike.

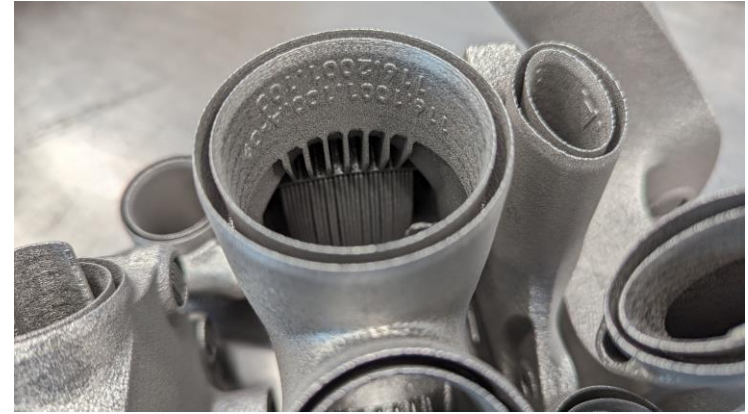
Concerns for Atherton Bikes in switching to coarser powder and thicker layers:

- Concerns include a reduction in surface quality that arise from the changes.
- High fatigue performance are important for these components.



Atherton
bikes

6. Demonstrator parts: Atherton Bikes

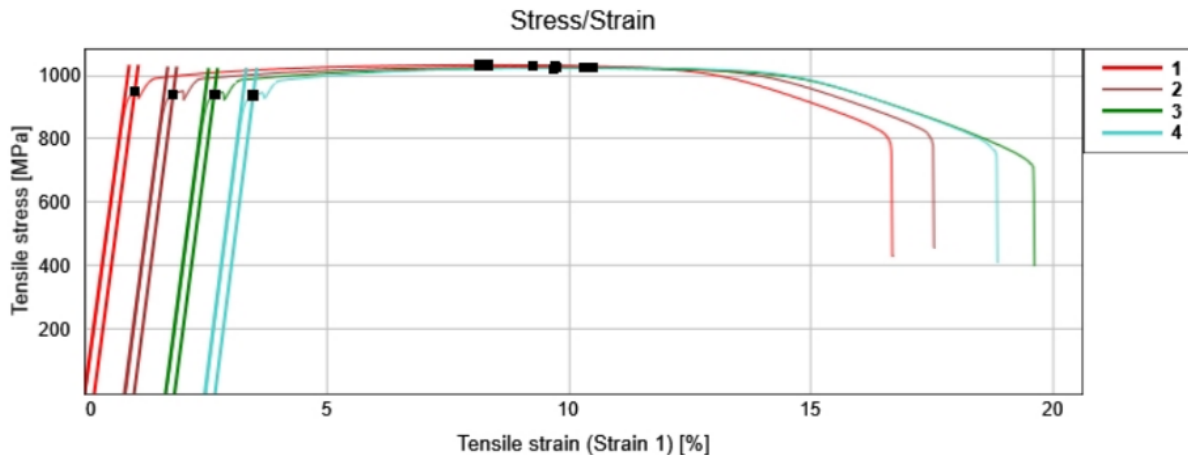


- Post processing includes supports removal and bead blasting of surface.
- Also printed the same layout in 60 μm layers for surface quality comparisons.
- Tensile bars printed on each plate (two from each build will be tested).
- Downskin surfaces appear rough on the more extreme overhang areas but labelling is clear and no issues with the supports observed.

Atherton
bikes

6. Demonstrator parts: Atherton Bikes - Results

- In order to examine the feasibility of Atherton Bikes switching their feedstock powder to the coarser range, two builds were carried out in the coarser powder (15-106 μm) in the standard parameters:
 - i. 90 μm layer thickness
 - ii. 60 μm layer thickness
- It is the standard procedure at Atherton Bikes to print four (4) tensile bars on each of their lug builds and two (2) from each layer thickness were tested under the same conditions previously described.
- The results were compared with results from the Renishaw data sheet.



Build conditions

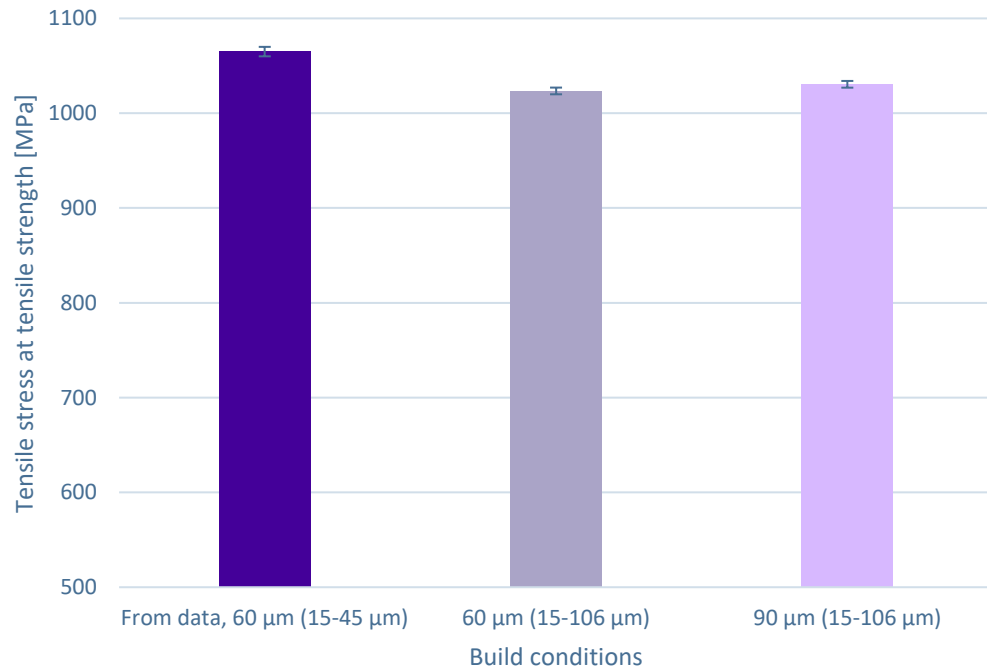
15-106 μm (90 μm)	15-106 μm (60 μm)
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1 & 2

3 & 4

6. Demonstrator parts: Atherton Bikes – Tensile strength results

Comparison of tensile stress at tensile strength for different build conditions

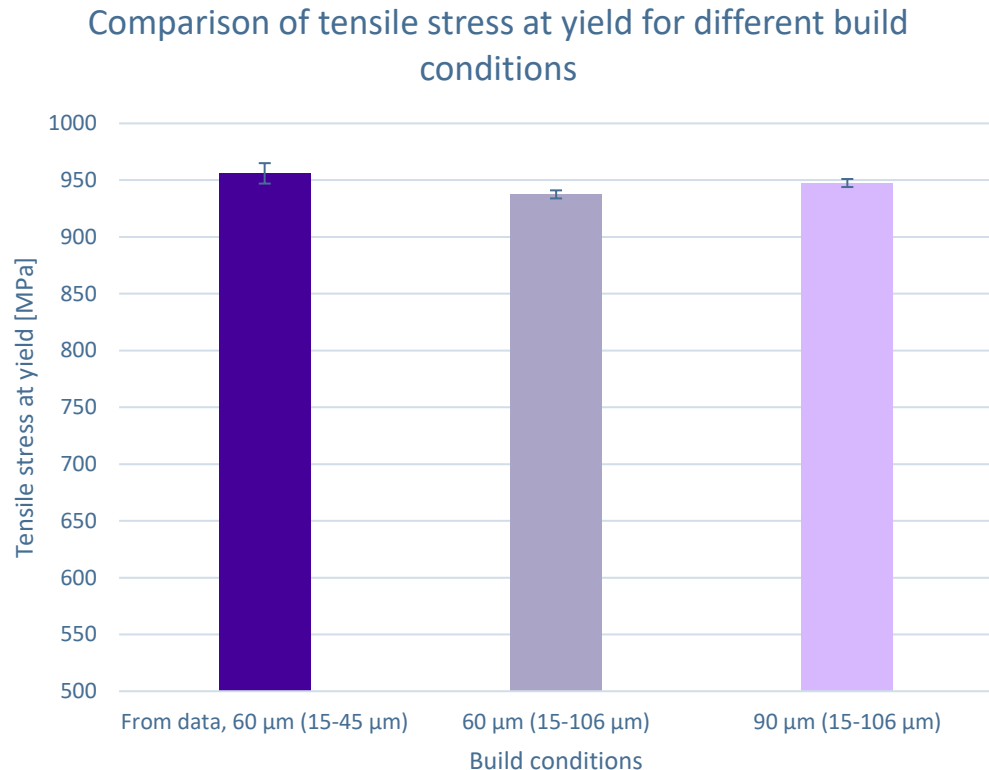


- There are two measurements from each build location on each specimen, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.
- It shows a minimal reduction in tensile strength when compared to the performance of specimens produced in the standard powder in standard layers (taken from the data sheet).

Build conditions

15-53 μm (60 μm)	15-106 μm (60 μm)	15-106 μm (90 μm)
1065 MPa	1024 MPa	1031 MPa

6. Demonstrator parts: Atherton Bikes – Yield stress results

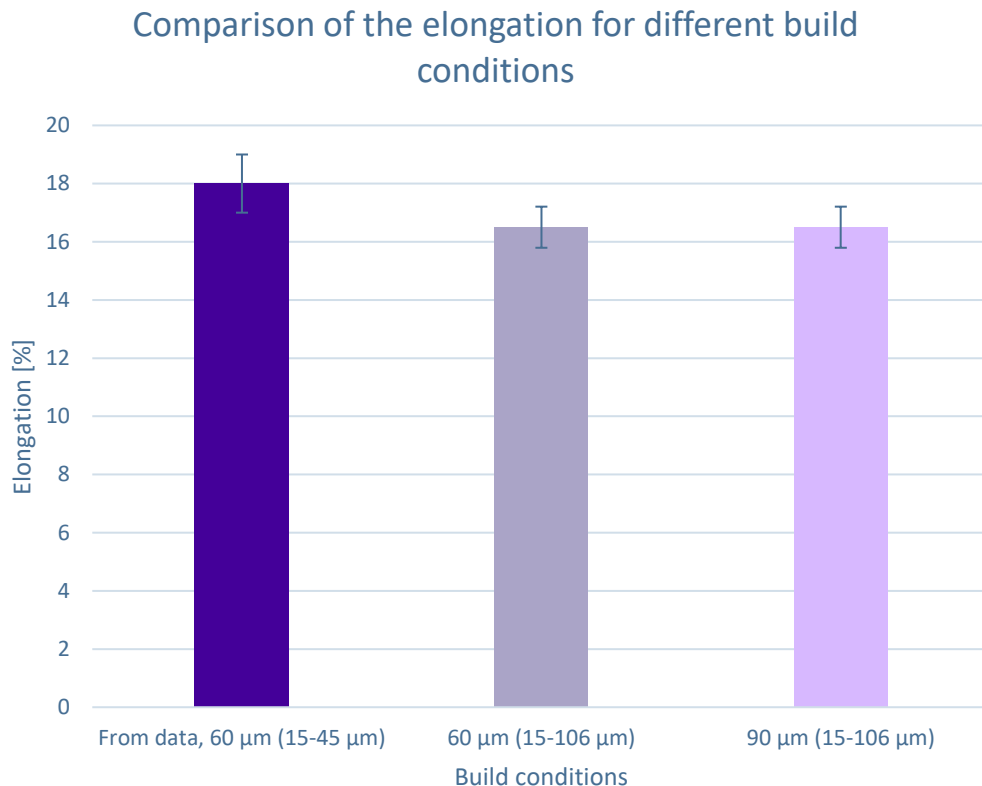


- There are two measurements from each build location on each specimen, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.
- It shows a minimal reduction in yield stress when compared to the performance of specimens produced in the standard powder in standard layers (taken from the data sheet).

Build conditions

15-53 μm (60 μm)	15-106 μm (60 μm)	15-106 μm (90 μm)
956 MPa	938 MPa	948 MPa

6. Demonstrator parts: Atherton Bikes – Elongation results



- There are two measurements from each build location on each specimen, with the standard deviation being calculated using the Excel Standard Deviation function and applied to give the error bars.
- It shows a minimal reduction in elongation when compared to the performance of specimens produced in the standard powder in standard layers (taken from the data sheet).

Build conditions		
15-53 μm (60 μm)	15-106 μm (60 μm)	15-106 μm (90 μm)
18 %	17 %	17 %

6. Demonstrator parts – observations and conclusions

- Whilst surface roughness is increased in the parts produced when compared to parts produced by Atherton Bikes in the standard (15-53 μm) powder, due to the post-processing and finishing of the parts, this may not be an issue for them.
- The mechanical properties of the specimens bars built alongside the lug components are comparable with the results from the standard parameters.
- Feedback from Atherton Bikes is positive and they are currently examining all the results and considering a switch to the coarser powder range for the future production of their parts as a result of this work.

Further work:

- An examination of the surface roughness of the parts produced in the coarser powder and the thicker layers. Quantification of the surface roughness in an as-built and finished condition. Can the contour laser parameters be optimised to improve the surface roughness?
- An analysis of the post-processing time – due to the increased surface roughness for the parts manufactured in the coarser powder (when compared with parts produced in the standard (15-53 μm) powder), will the finishing of the lug components result in an increased finishing process? This is an analysis that will be conducted by the team at Atherton Bikes.

7. Conclusions and further work

7. Conclusions



By switching the feedstock powder to the coarser range, there is a cost saving of between 25 and 30 %*

* Cost estimate provided by Carpenter Additive UK



Build times can be reduced by a third when printing in a layer thickness of 90 μm , (when compared with a 60 μm layer thickness).



Density and mechanical properties for the coarser powder and the thicker layers are comparable with results from the recommended powder size range and layer thickness.

7. Conclusions

- 1.** This project aimed to fill the gap in the knowledge as to whether a coarser range of powder (15-106 μm) could be deployed in a production AM machine in order to reduce costs, without reducing manufactured part quality.
- 2.** An examination of the density of specimens produced in the coarser powder (15-106 μm) at a layer thickness of 120 μm demonstrated that very low porosity and faster build rates could be achieved, however, due to increased surface roughness and feedback from our partners at Renishaw (they had initial results of a reduction in the mechanical performance of the specimens they produced), the focus was switched to the development of a set of parameters to build 90 μm layers.
- 3.** High density was achieved when building in the standard powder (15-53 μm). Across the build plate, density varied between 99.704 % to 99.958 %
- 4.**
 - Excellent density was obtained in both 60 μm (99.973 %) and 90 μm (99.963 %) when building in the coarser powder (15-106 μm).
 - Over the course of 10 builds of differing z heights, a slight shift in the PSD range has been observed, however, this is without the more typical addition of virgin powder to the system.

7. Conclusions

- The RenAM 500Q machine, with the in-built integrated sieve has performed well with no issues coping with the increased particle sizes. Flowability is better with the coarser powder (15-106 μm) than with the standard powder (15-53 μm) in both the virgin and the used state.
- There was no significant increase in the content of oxygen, (Build 1 = 0.084 % and build 10 = 0.086 %) over the course of the 10 builds carried out in the coarser powder.
- The microhardness results for specimens produced in the coarser powder were consistent despite the different layer thickness: 357.7 HV 0.1 (60 μm layers) and 357.3 HV 0.1 (90 μm layers) on average.
- Build location has an impact on the mechanical performance of the specimens produced, with the left side of the build and specifically the left-bottom area performing the worst, however, the results for tensile strength, yield stress and elongation are not significantly different from the results of specimens built in the standard powder (15-53 μm) and in the standard layer thickness (60 μm) – data taken from the Renishaw data sheet.
- At 50 % of the yield stress, the bottom left specimens for both 60 and 90 μm layer thicknesses test bars failed before 100,000 cycles and at 50 % of the yield stress, the top right specimens for both 60 and 90 μm layer thicknesses test bars did not fail and continued past 1,000,000 cycles, indicating that build location has an influence on the fatigue properties, over differences in layer thickness.

7. Conclusions

- It is possible that variability in the gas flow in different locations is having an impact on the mechanical properties, however, further investigations are required. The readings from the anemometer show that the bottom-left area of the build platform experiences the highest gas flow rate when compared to the other bed locations and it is in this area that tensile and fatigue performance is at it's lowest.

5. An attempt to reduce the volatility of the melt area by reducing the power and scan speed failed to improve the mechanical properties of the specimens produced. This method increased the build time and therefore does not make economical sense to pursue without the improvements to the mechanical performance.

6. Due to the reduced cost of the coarser feedstock powder and with the mechanical properties of the specimens bars built alongside the lug components being comparable with the results from the standard parameters, Atherton Bikes are switching to the coarser powder range for the future production of their parts as a result of this work.

7. Further work

- Further builds would be required to get an impression of the long term use of this coarser powder to make any firm conclusions of what the impact of repeated use of this powder would be on the PSD range.
- A closer examination to determine the effects of an increased flowability on the dosing quantities and a greater packing density, looking specifically at the affect on the thermal gradients, cooling rates and the resulting microstructure.
- Enhanced tensile strength and microhardness are the result of a finer grain structure, but will result in poorer ductility as hardness of Ti-6Al-4V is dependent on the grain size, specifically the thickness of the α laths. Further analysis of the microstructure could show the reason for the higher hardness measurements in the standard powder (15-53 μm) in the Coarser layers (90 μm).
- For a true comparison of specimens between the standard and coarser powder ranges, a build in the standard powder range (15-53 μm) could be built and tested, rather than comparing data from the Renishaw data sheet.
- Fracture analysis of the cause of the breaks, looking specifically for any initiation sites, porosity or inclusions that could indicate the cause of the failures.

7. Further work

- Further investigations on the significance of a varied gas flow across the bed could be carried out to determine to what extent whether gas flow irregularities contribute to differences in mechanical properties.
- An examination of the microstructure using the SEM and how it can be altered through laser process or alternative heat treatments. This can be related back to the mechanical results to determine the relationship between the microstructure and the mechanical performance.
- For the Atherton Bikes lug components: Quantification of the surface roughness of the parts produced in the coarser powder and the thicker layers and comparing the as-built and finished condition will be carried out to determine which layer thickness Atherton Bikes are willing to manufacture in. Further work to optimise the contour laser parameters to improve the surface roughness.
- For the Atherton Bikes lug components: An analysis of the post-processing time – due to the increased surface roughness for the parts manufactured in the coarser powder (when compared with parts produced in the standard (15-53 μm) powder), will the finishing of the lug components result in an increased finishing process? This is an analysis that will be conducted by the team at Atherton Bikes.

With thanks to:

Nick Weeks and Joseph Oluleke from Carpenter Additive UK



Ravi Aswathanarayanaswamy from Renishaw UK



Scott Aisthorpe and Dan Brown from Atherton Bikes



Thank you.

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